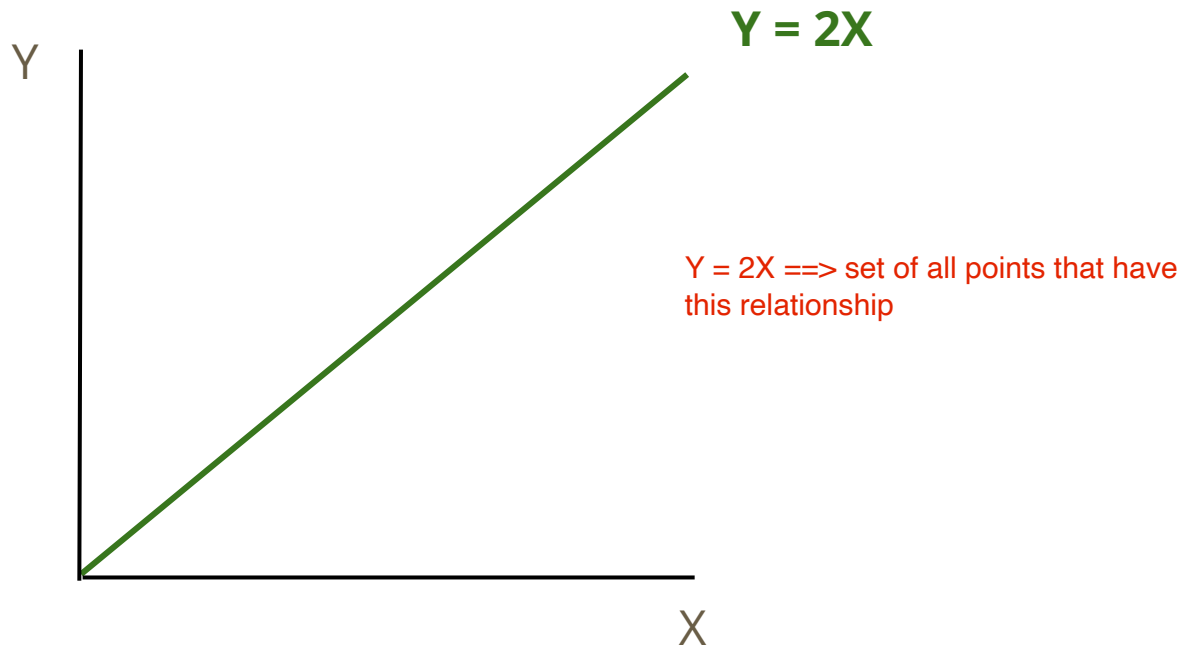
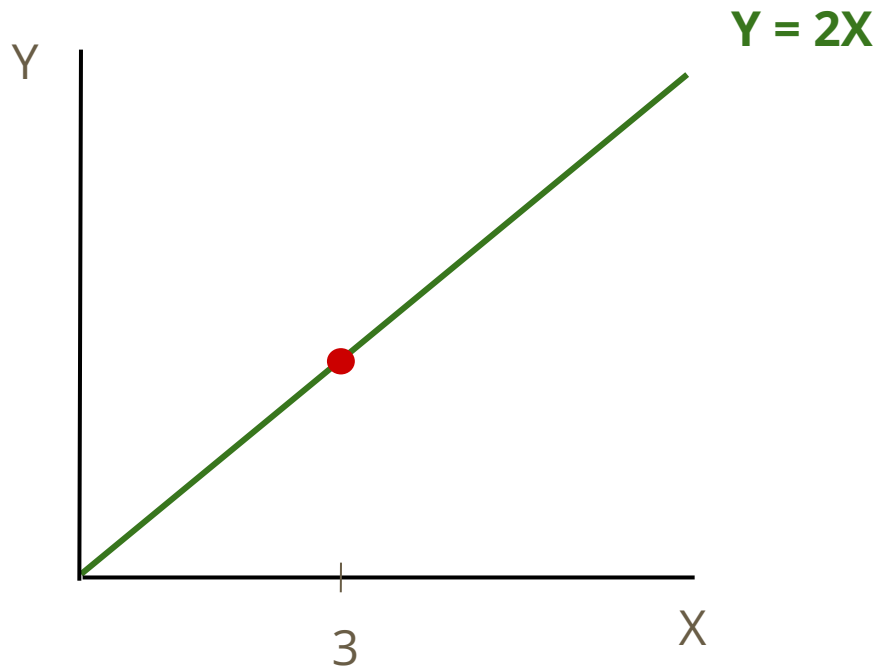

Linear Model Evaluation

— Boston University CS 506 - Lance Galletti —

Linear Function

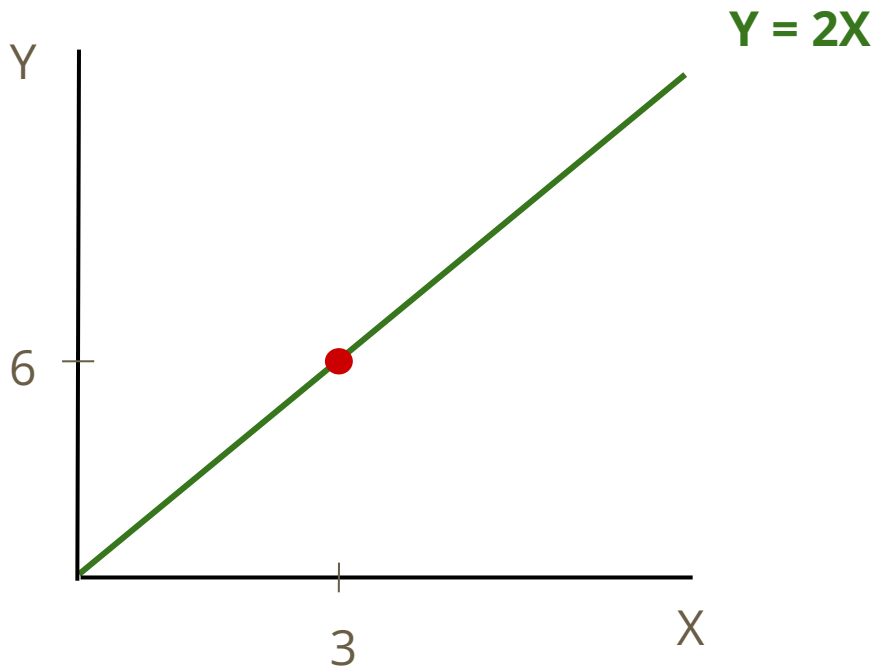


Linear Function

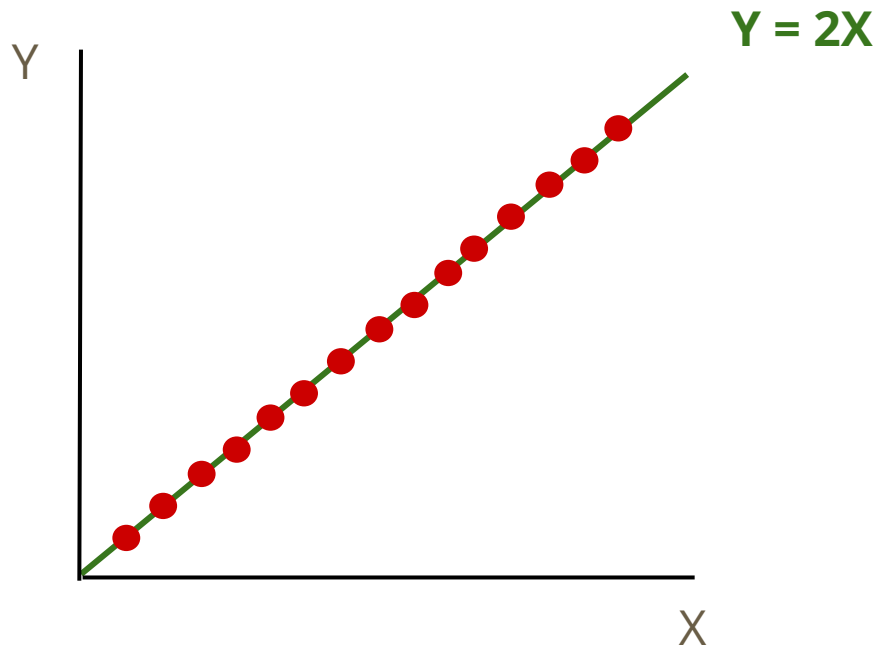


Linear Function

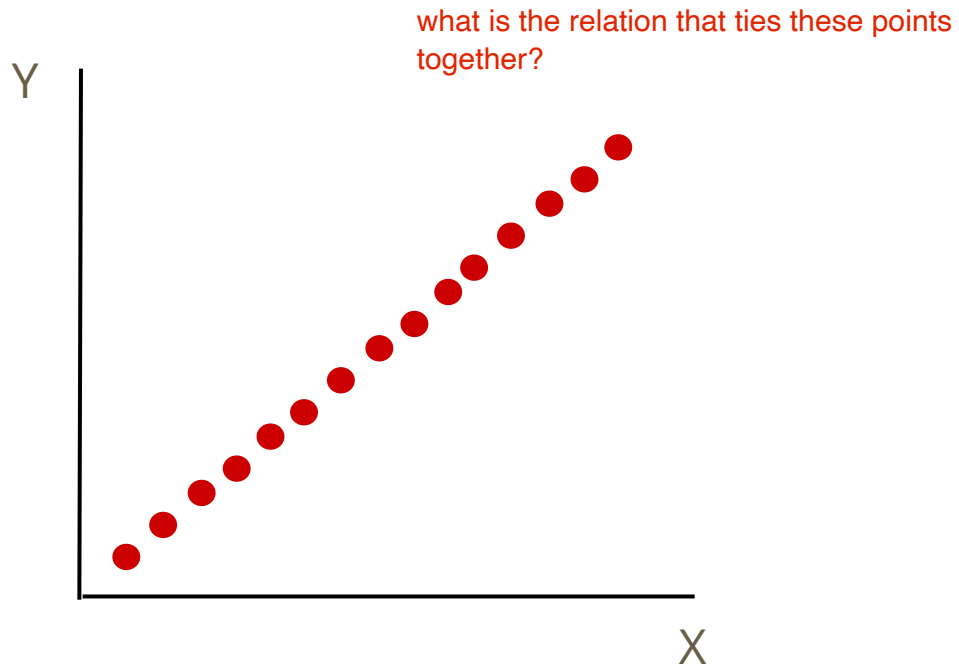
I know for sure what
value of Y I'm gonna get



Linear Function

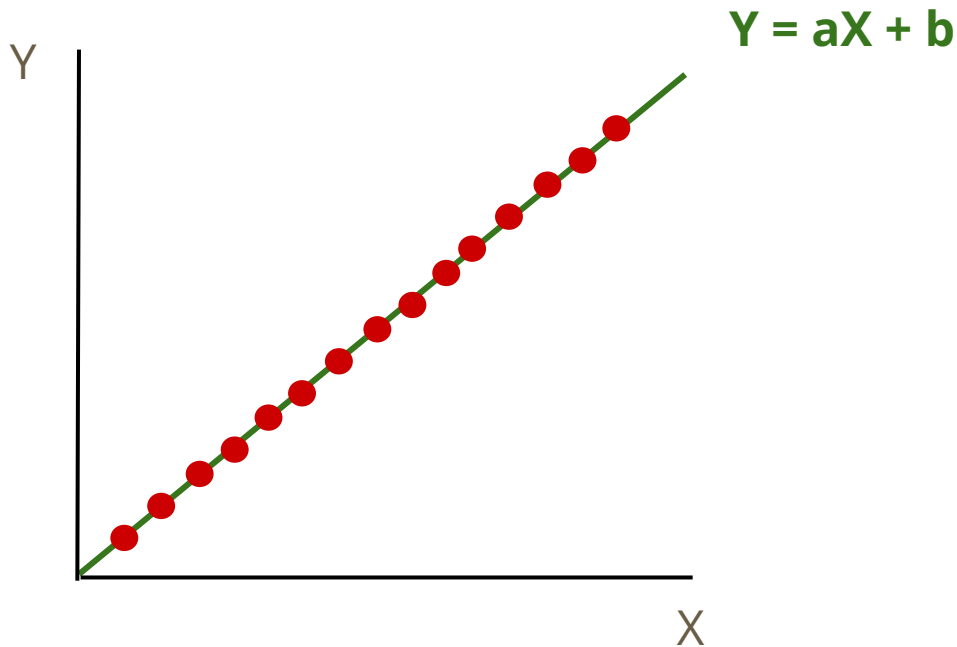


Ideally

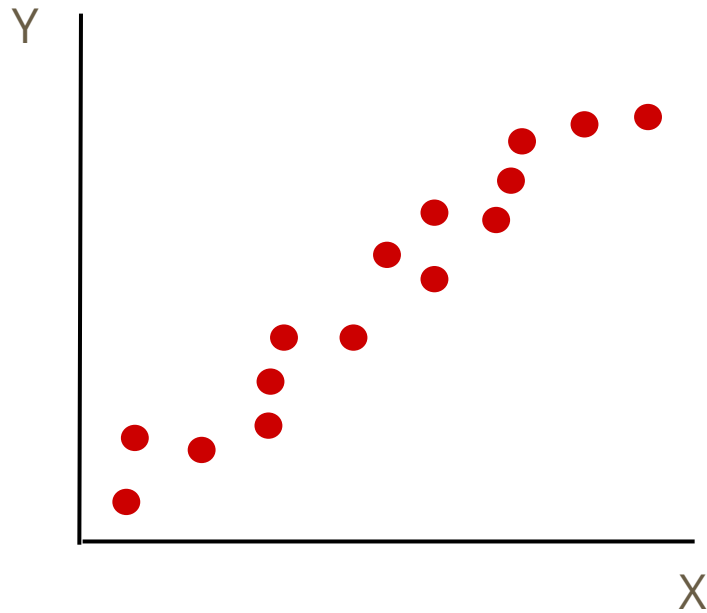


Ideally

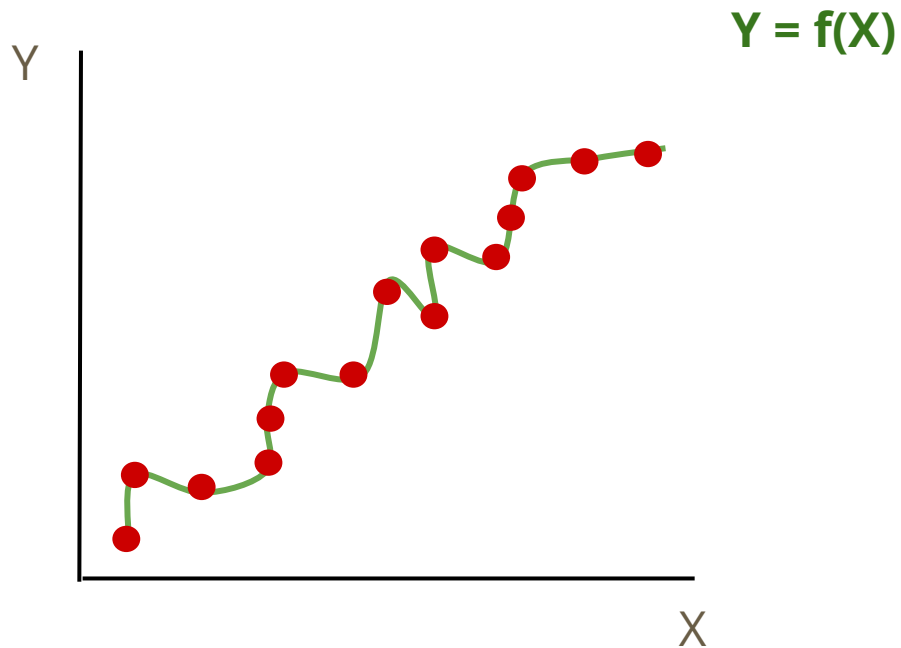
Guess the relationship
(**a** and **b**) from the data



Practically



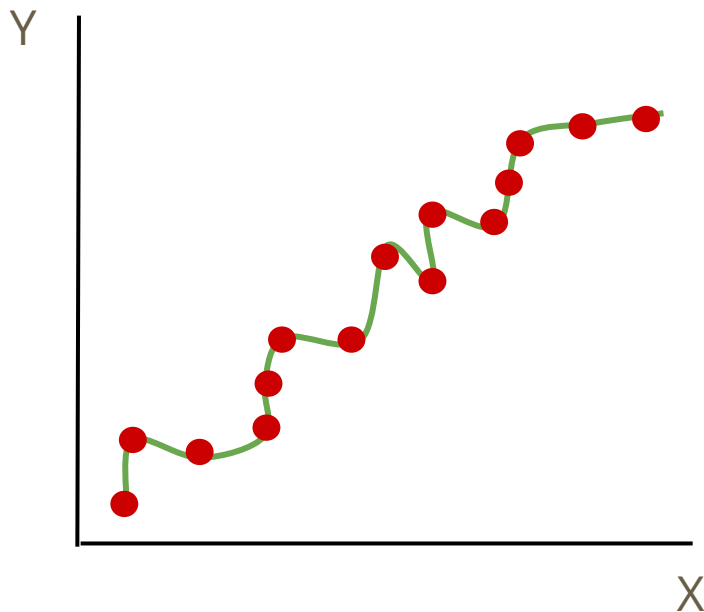
Practically



Practically

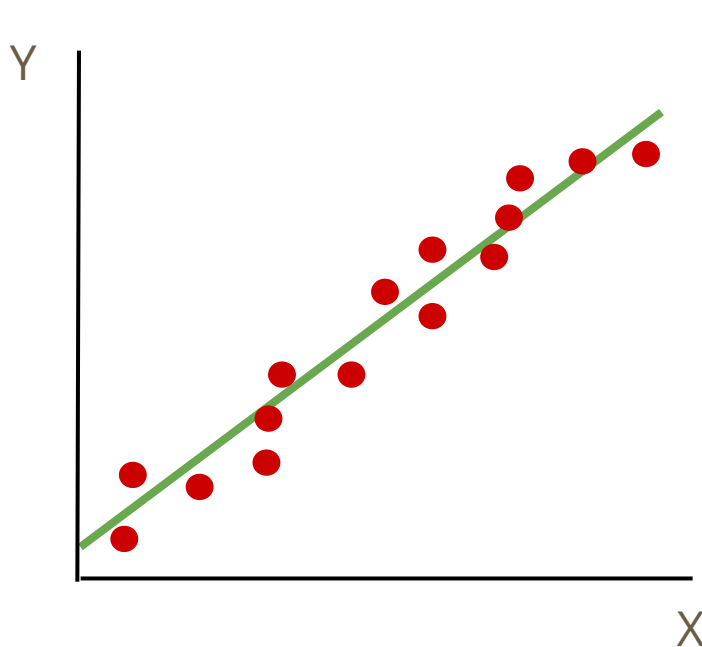
not a solvable problem (realistically)

$$Y = f(X)$$



f can be anything: too complex a problem to solve...

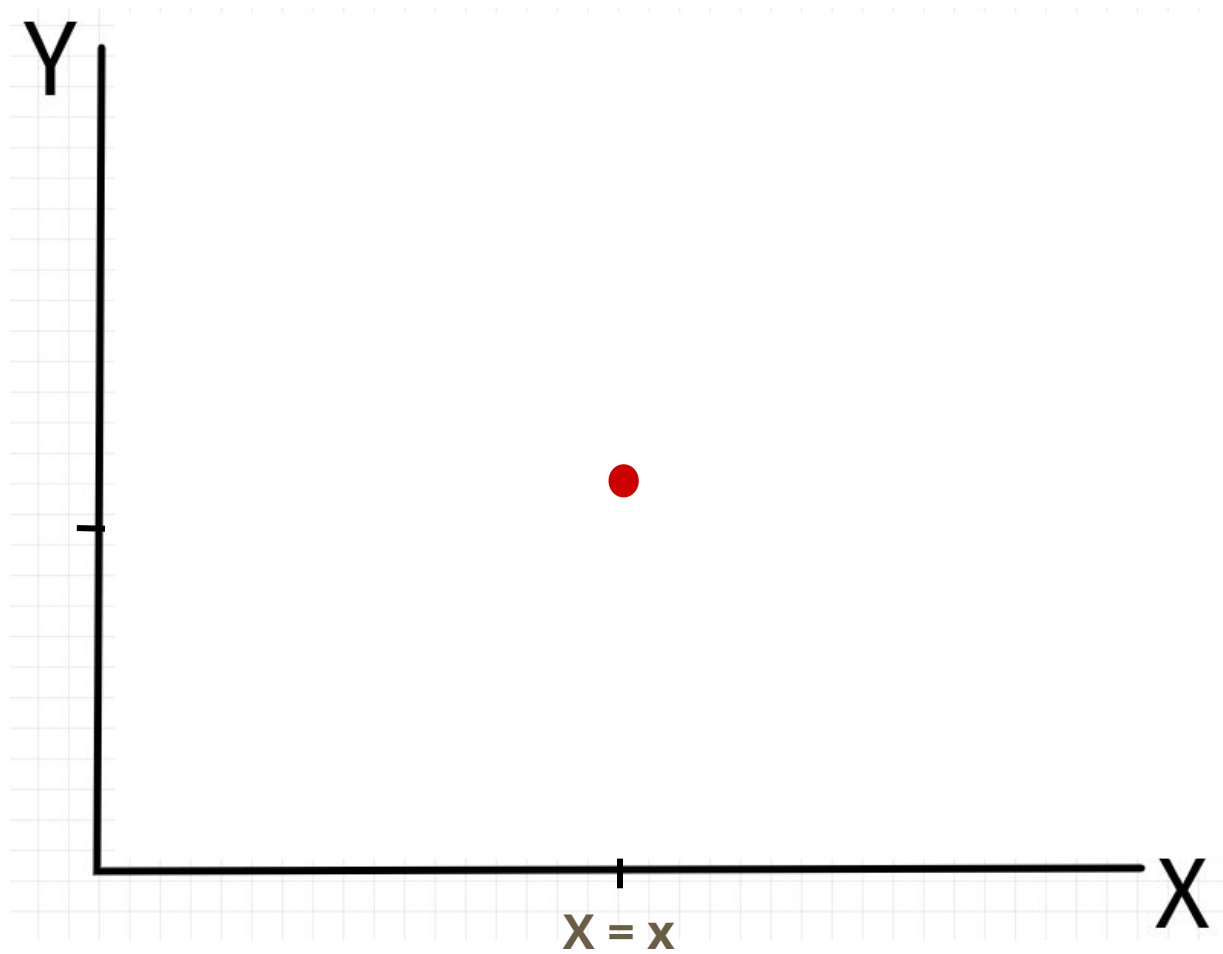
Practically



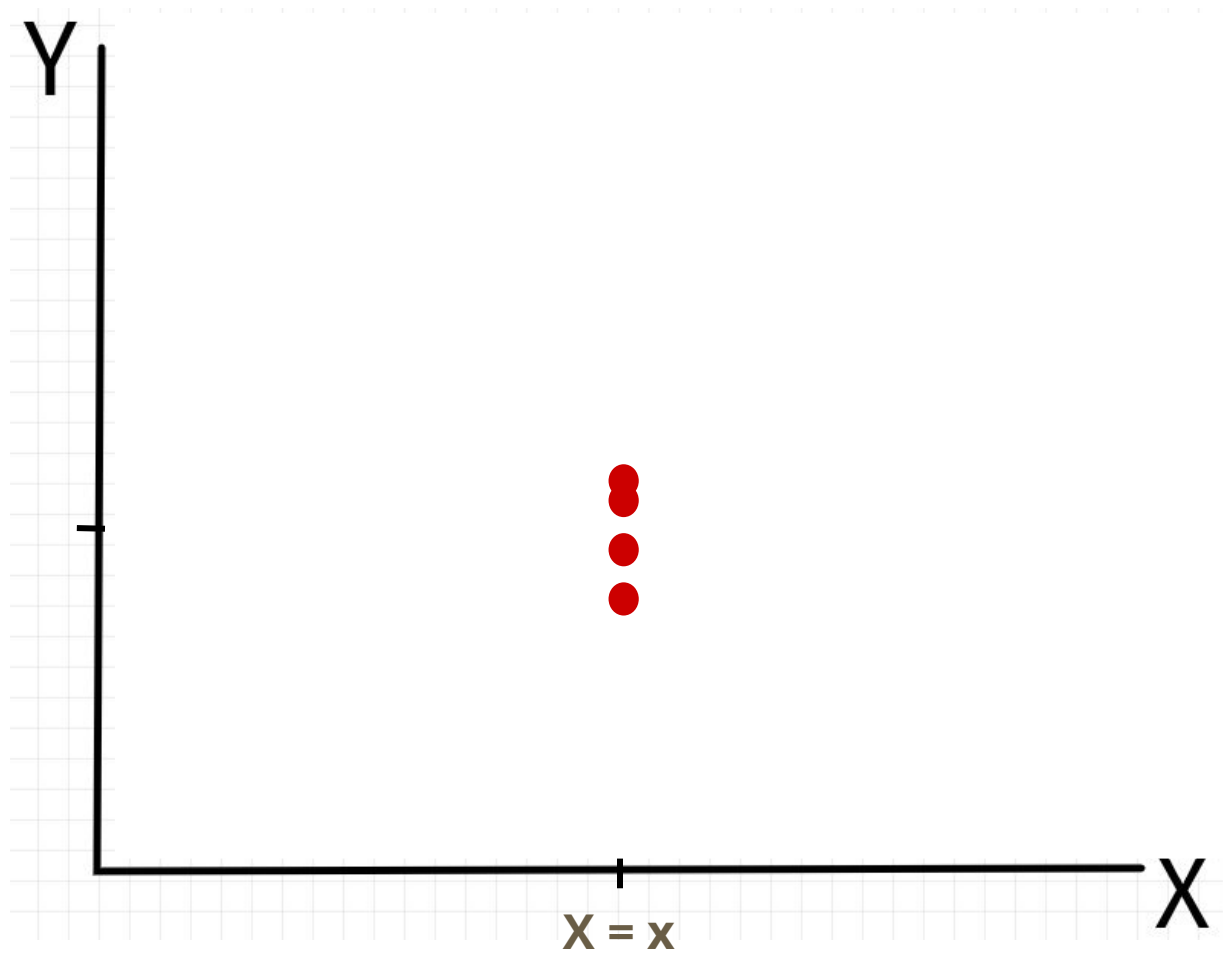
$$Y = f(X) + \text{noise}$$

Assume f is linear and the variation we're seeing is noise

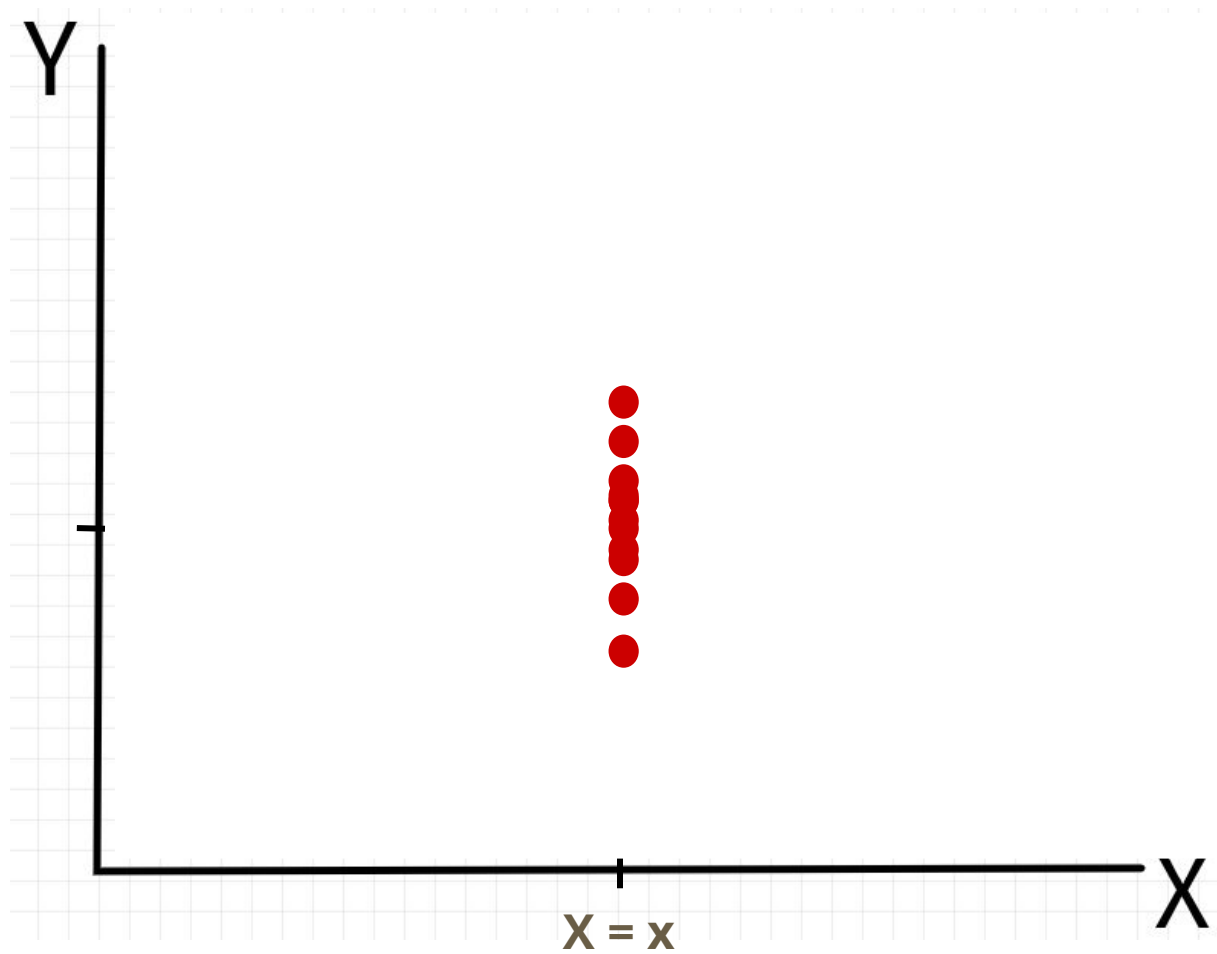
Assumptions



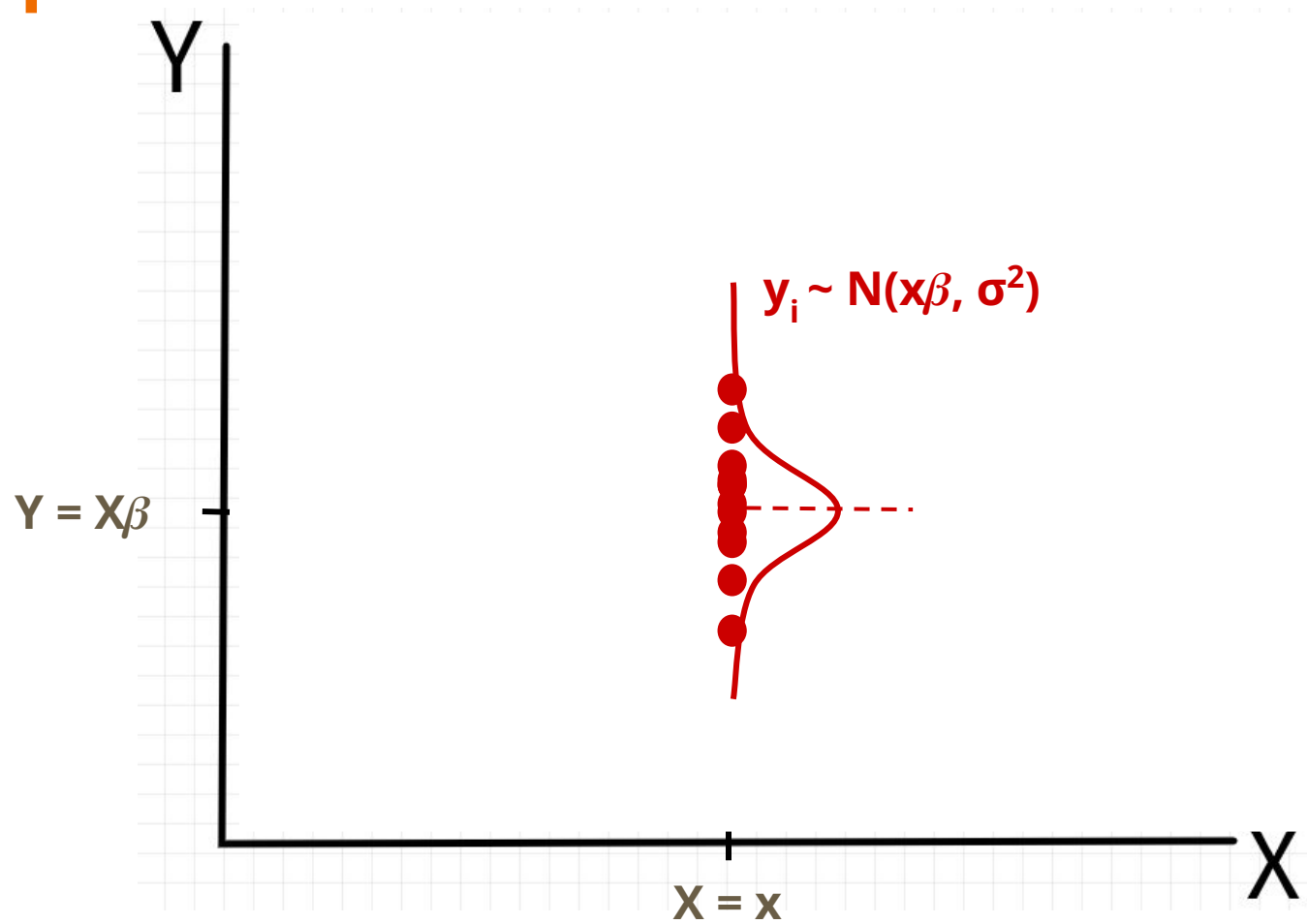
Assumptions



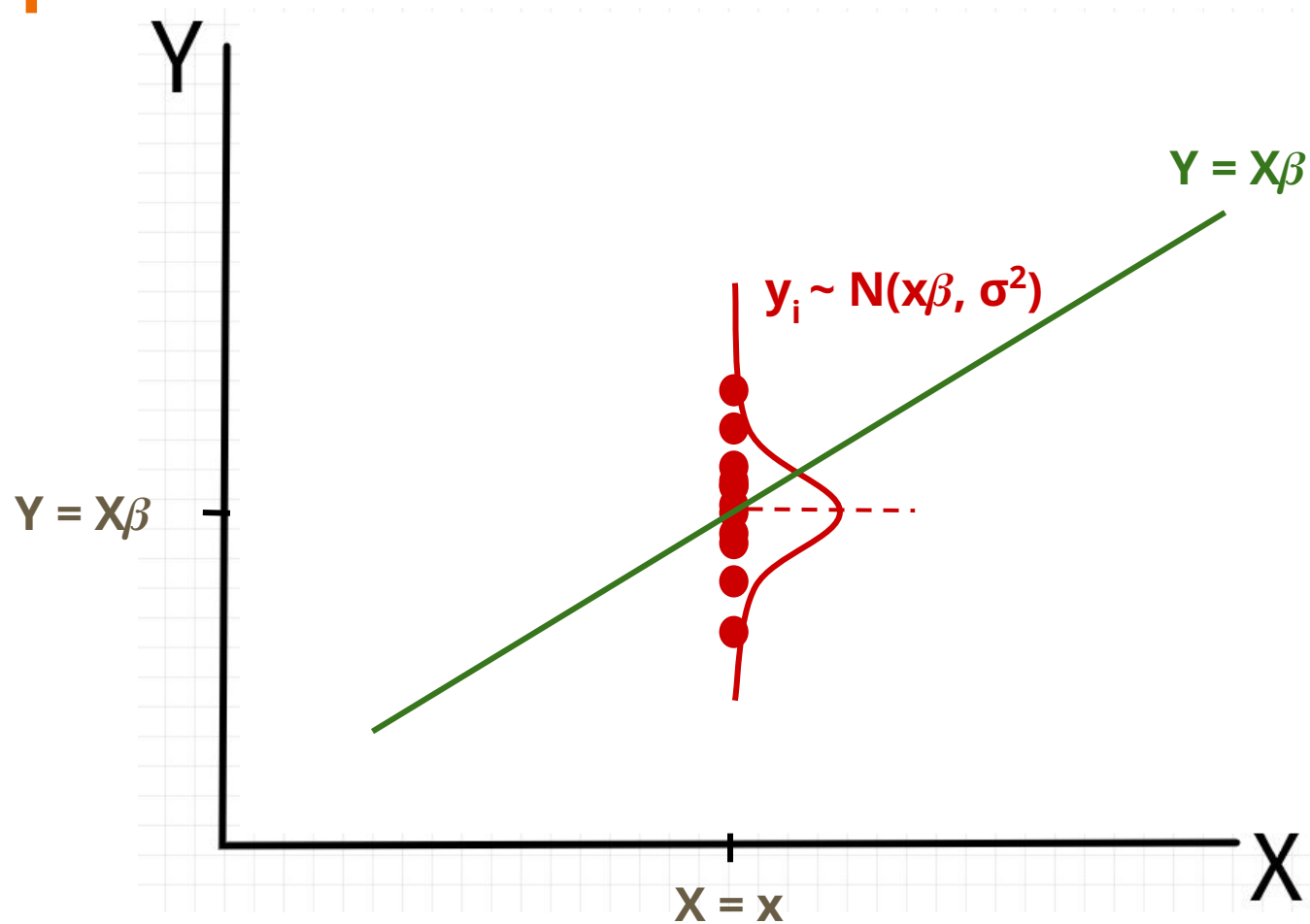
Assumptions



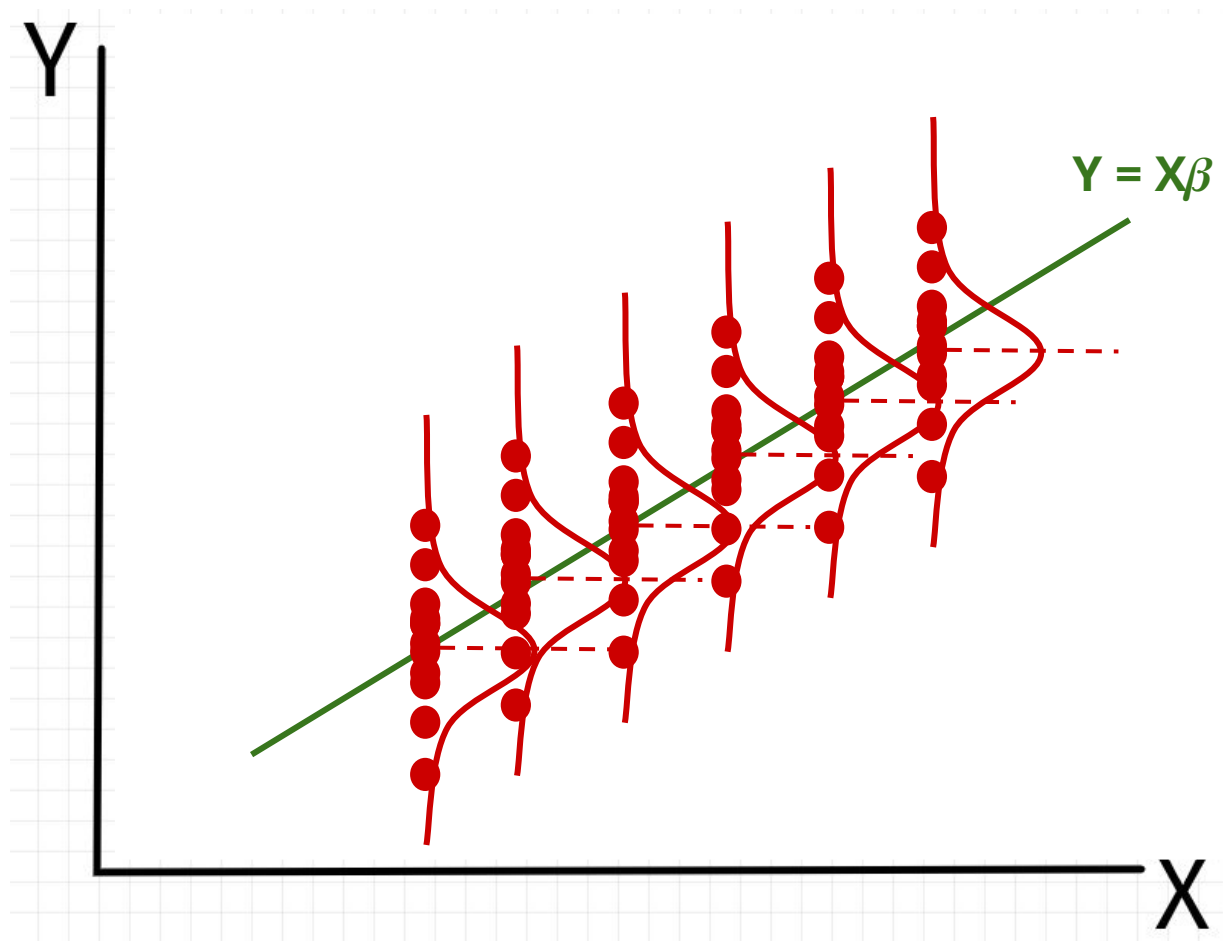
Assumptions



Assumptions



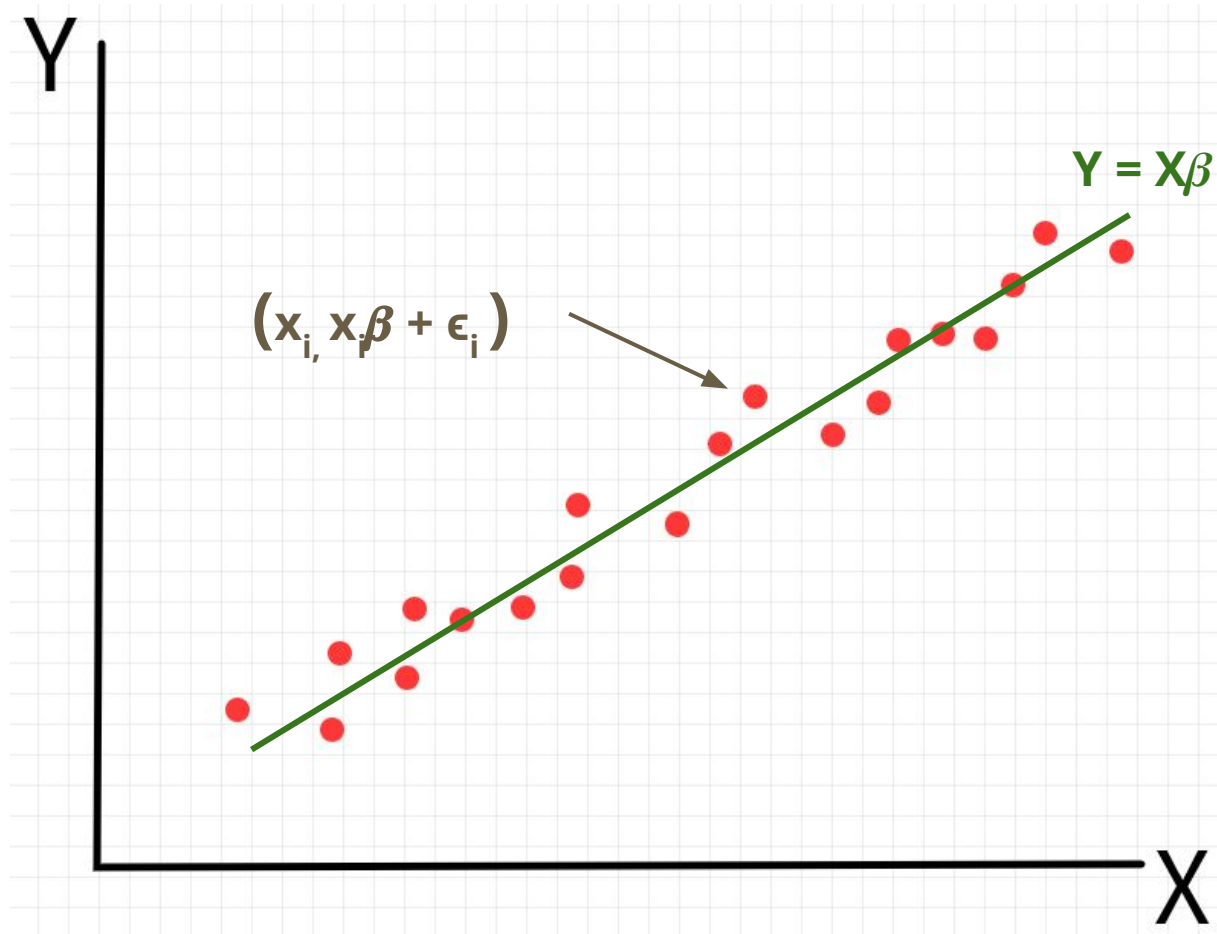
Assumptions



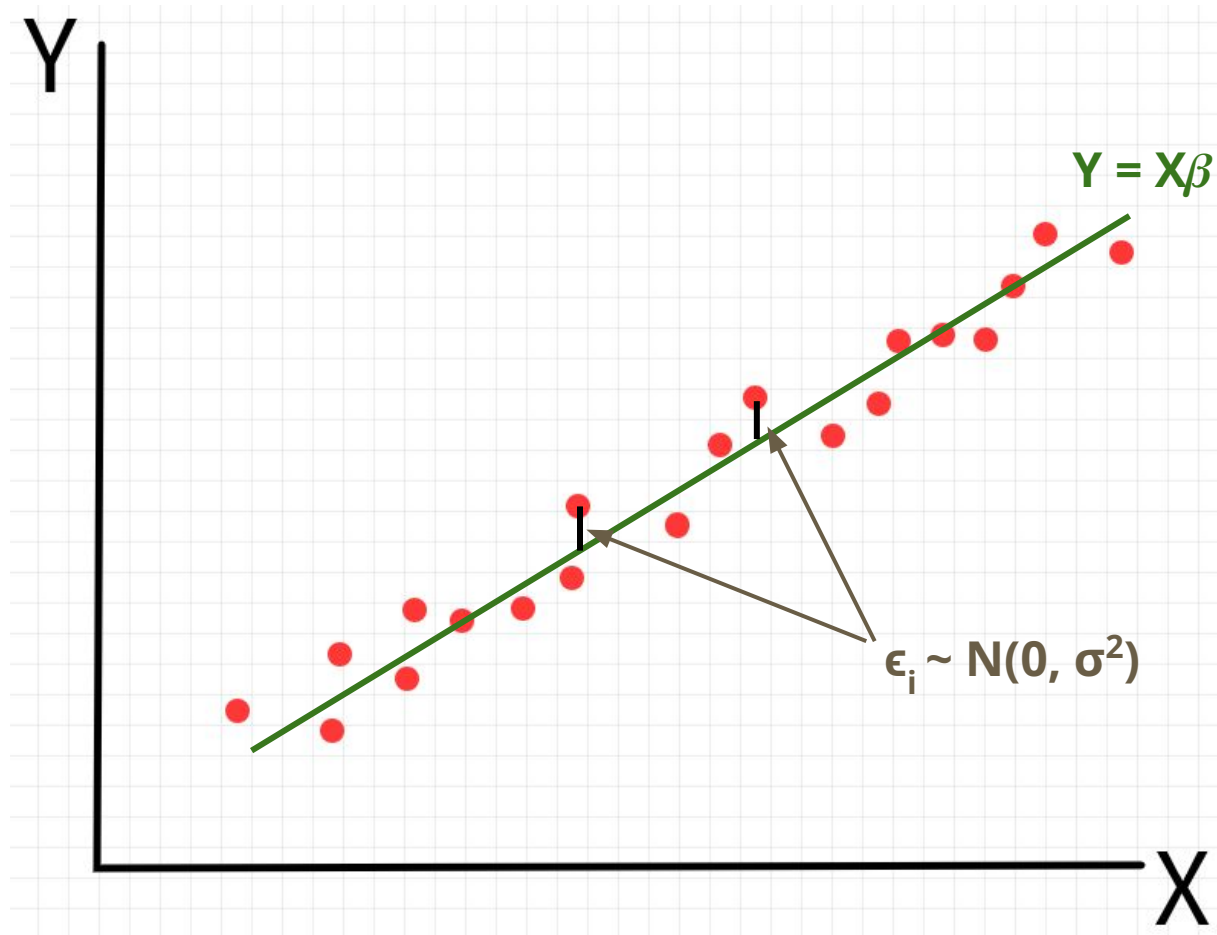
Assumptions



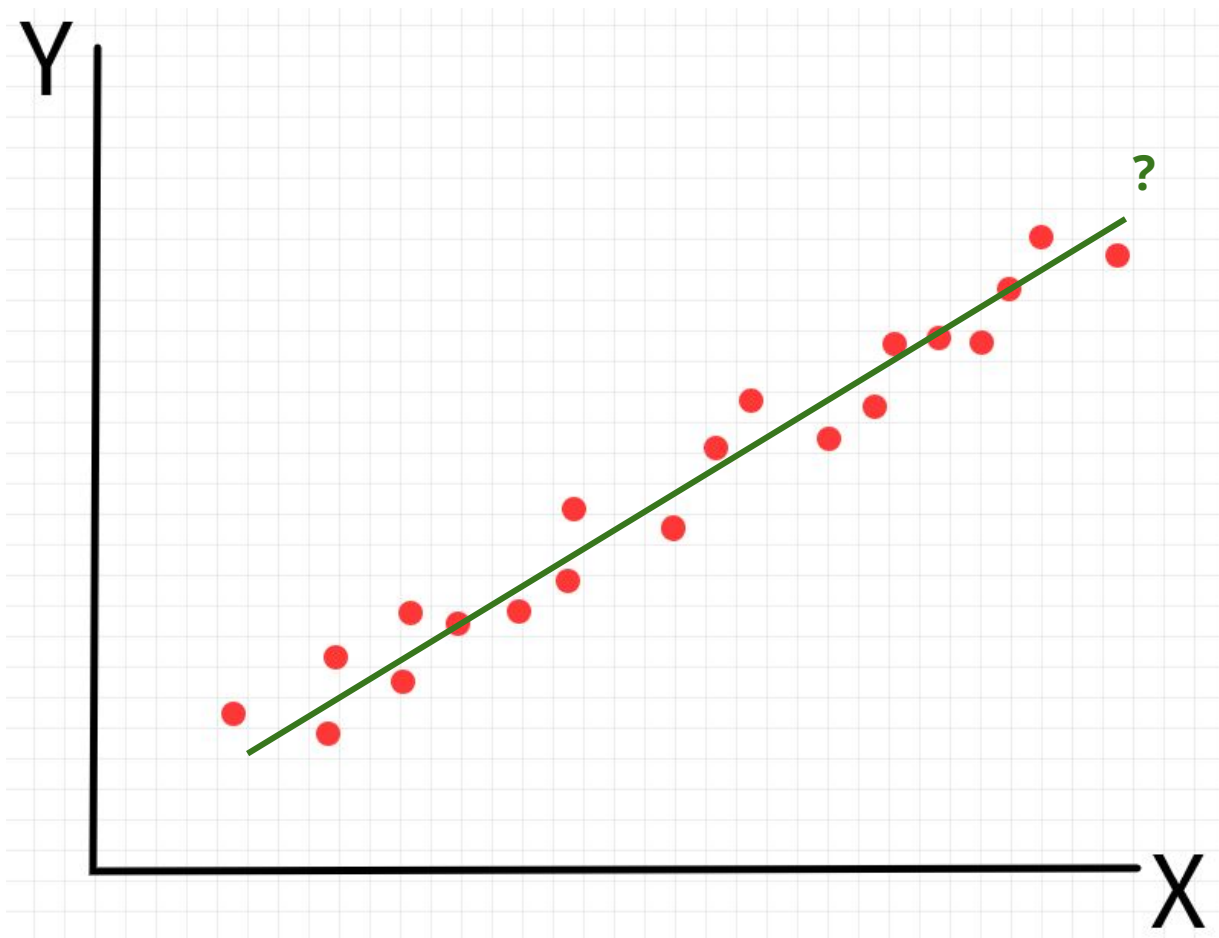
Assumptions



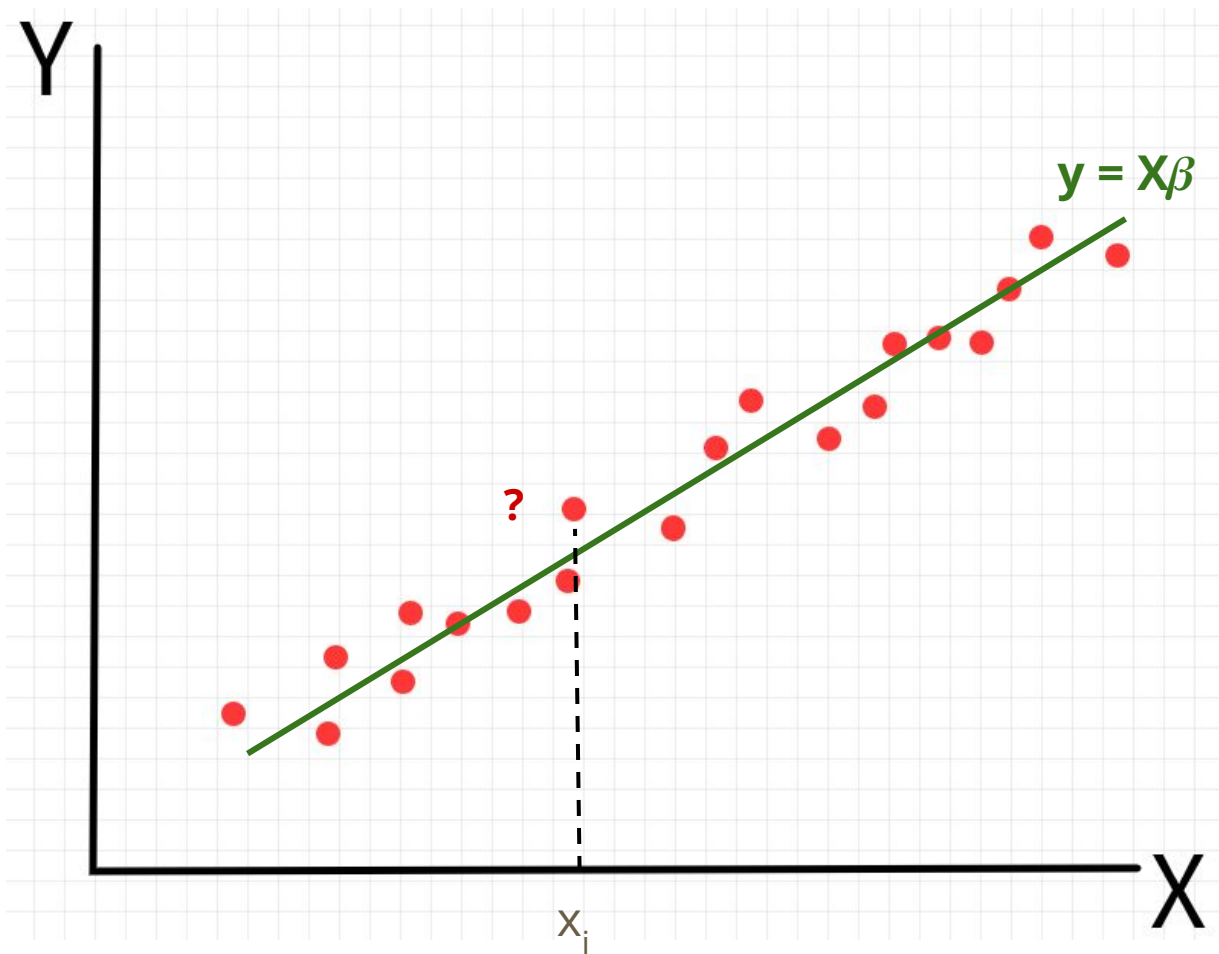
Assumptions

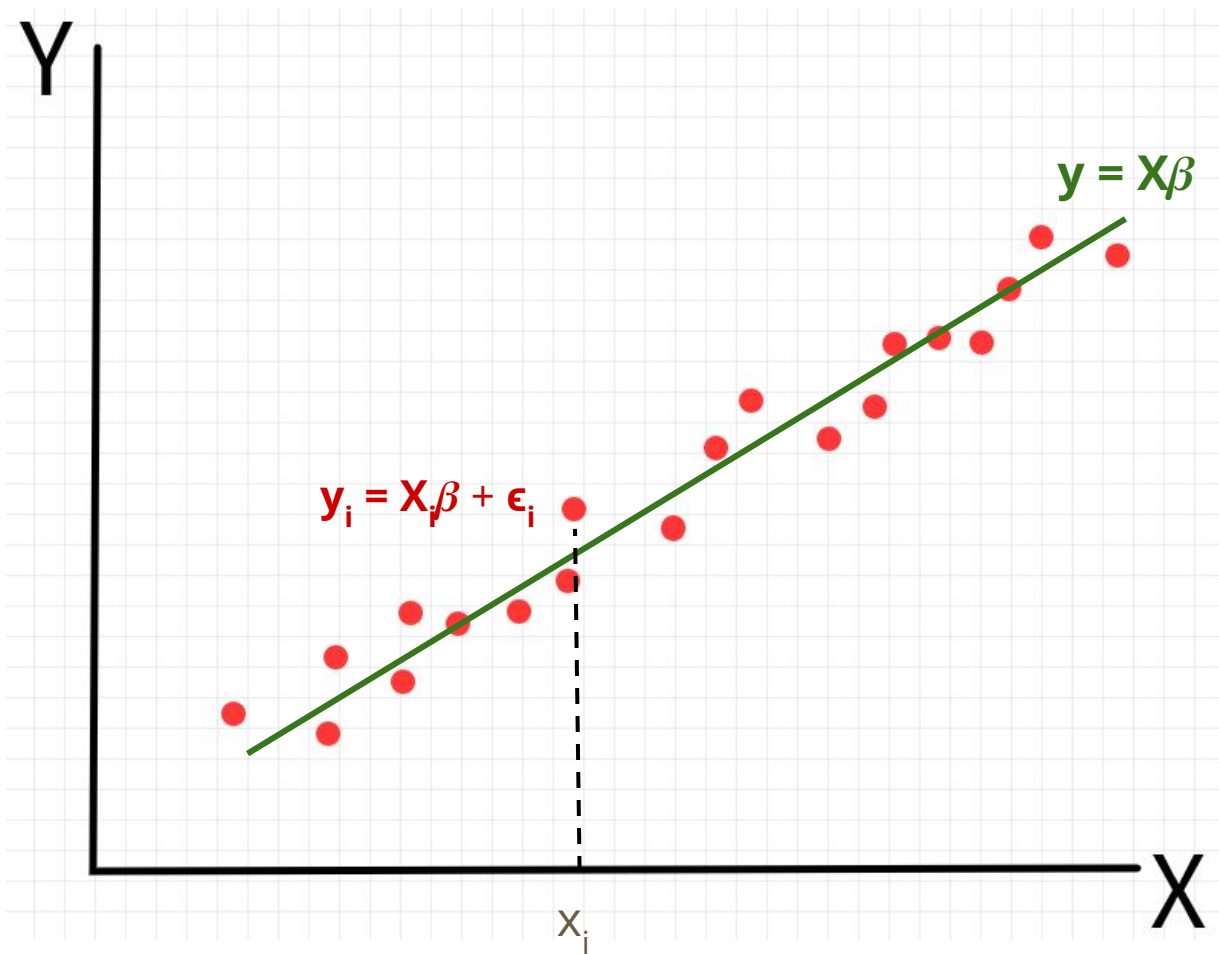


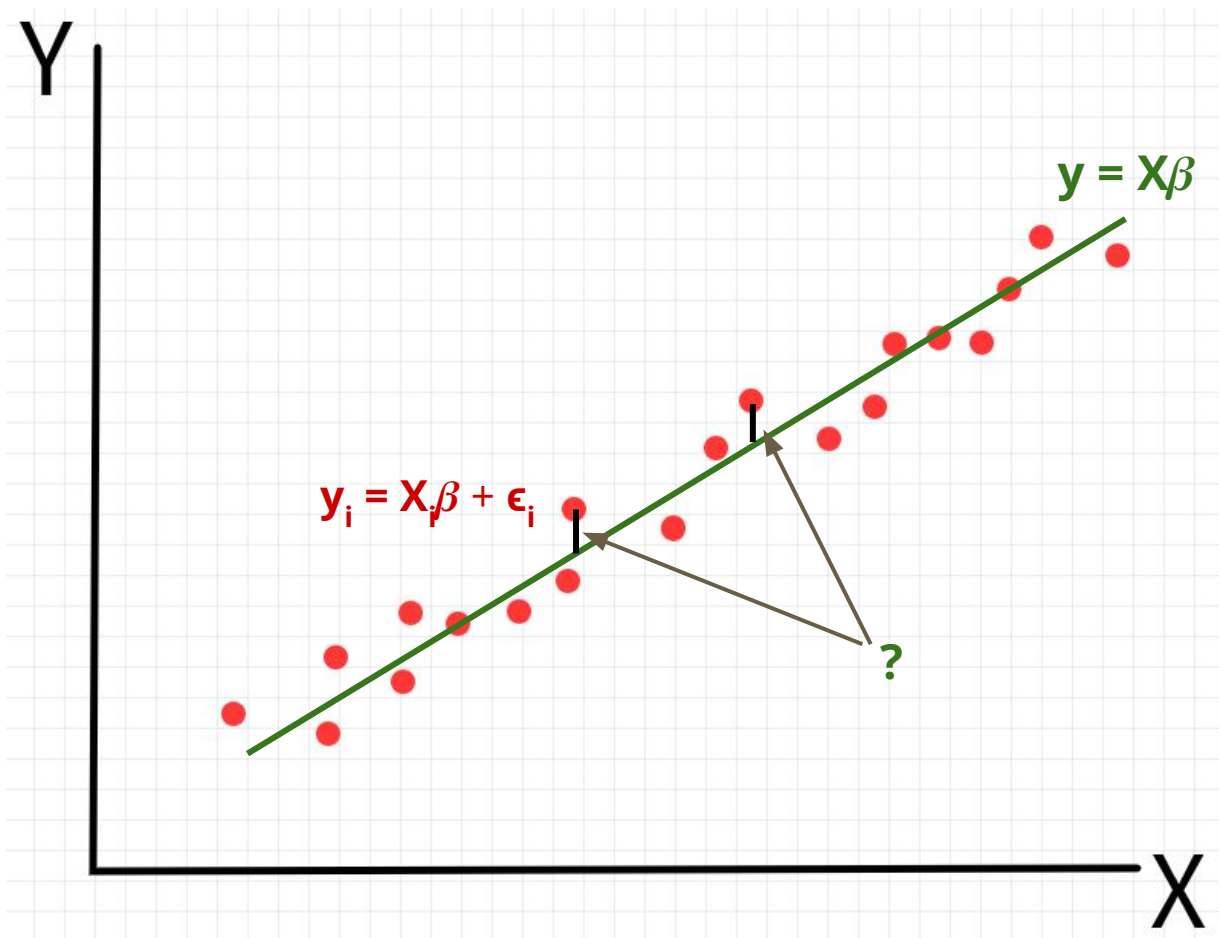
Let's label the diagram

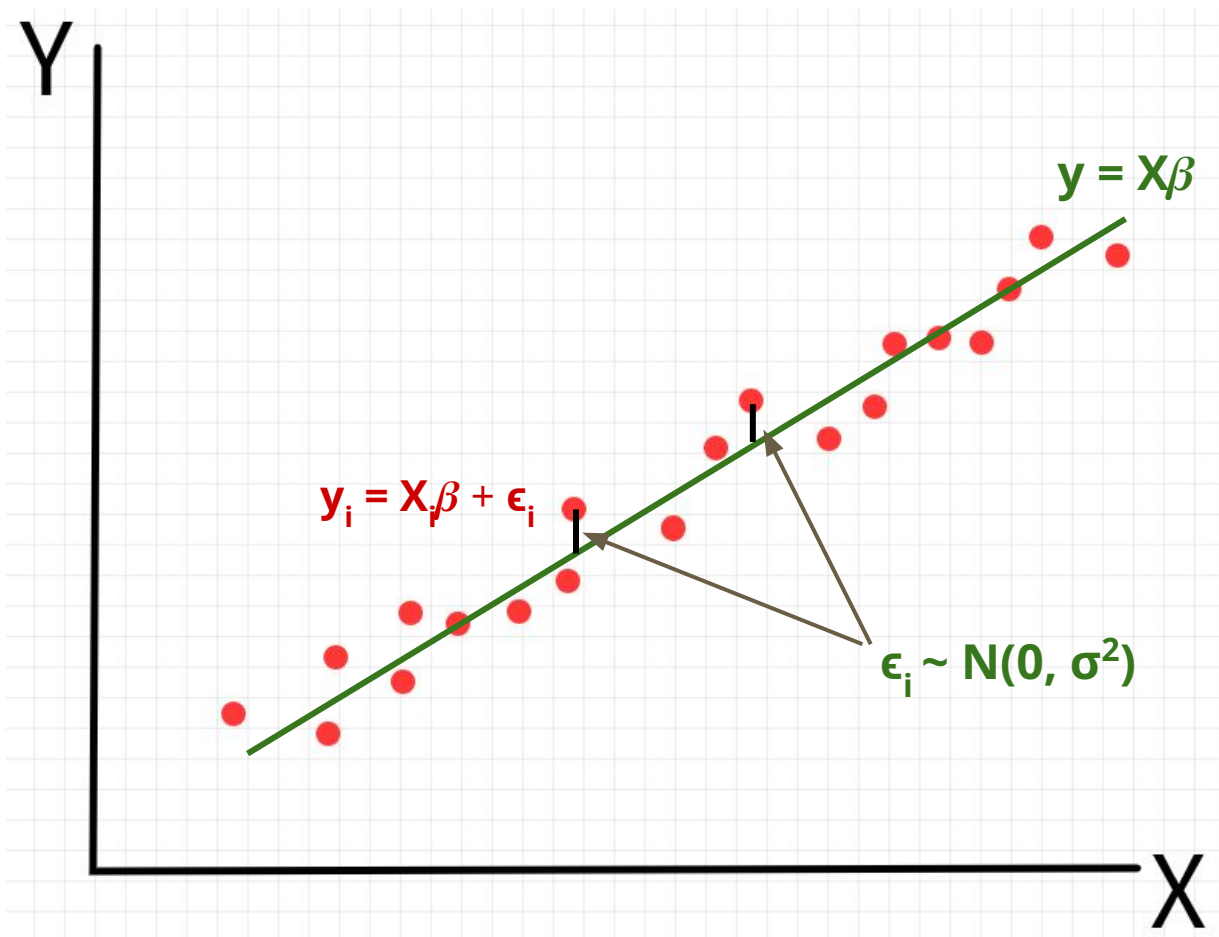


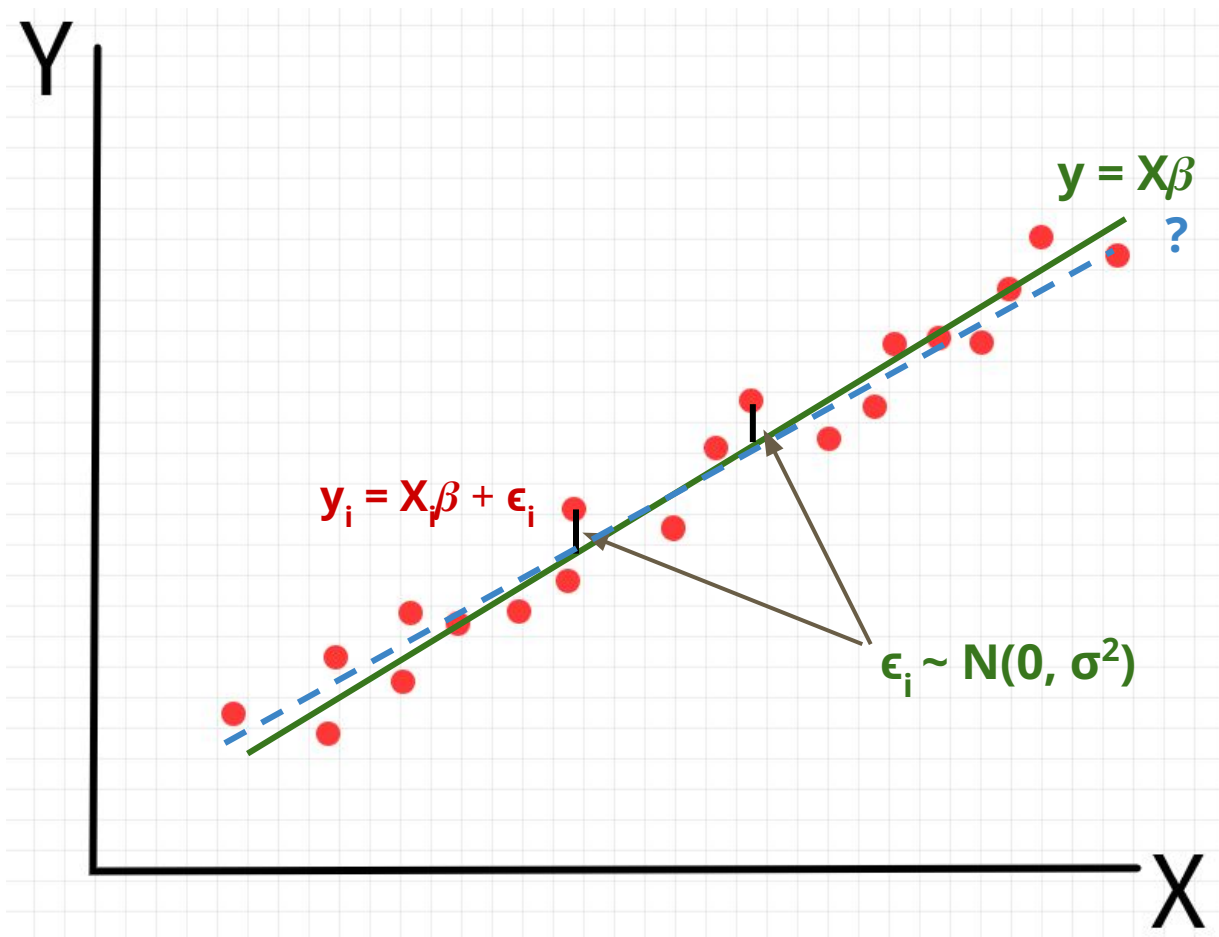


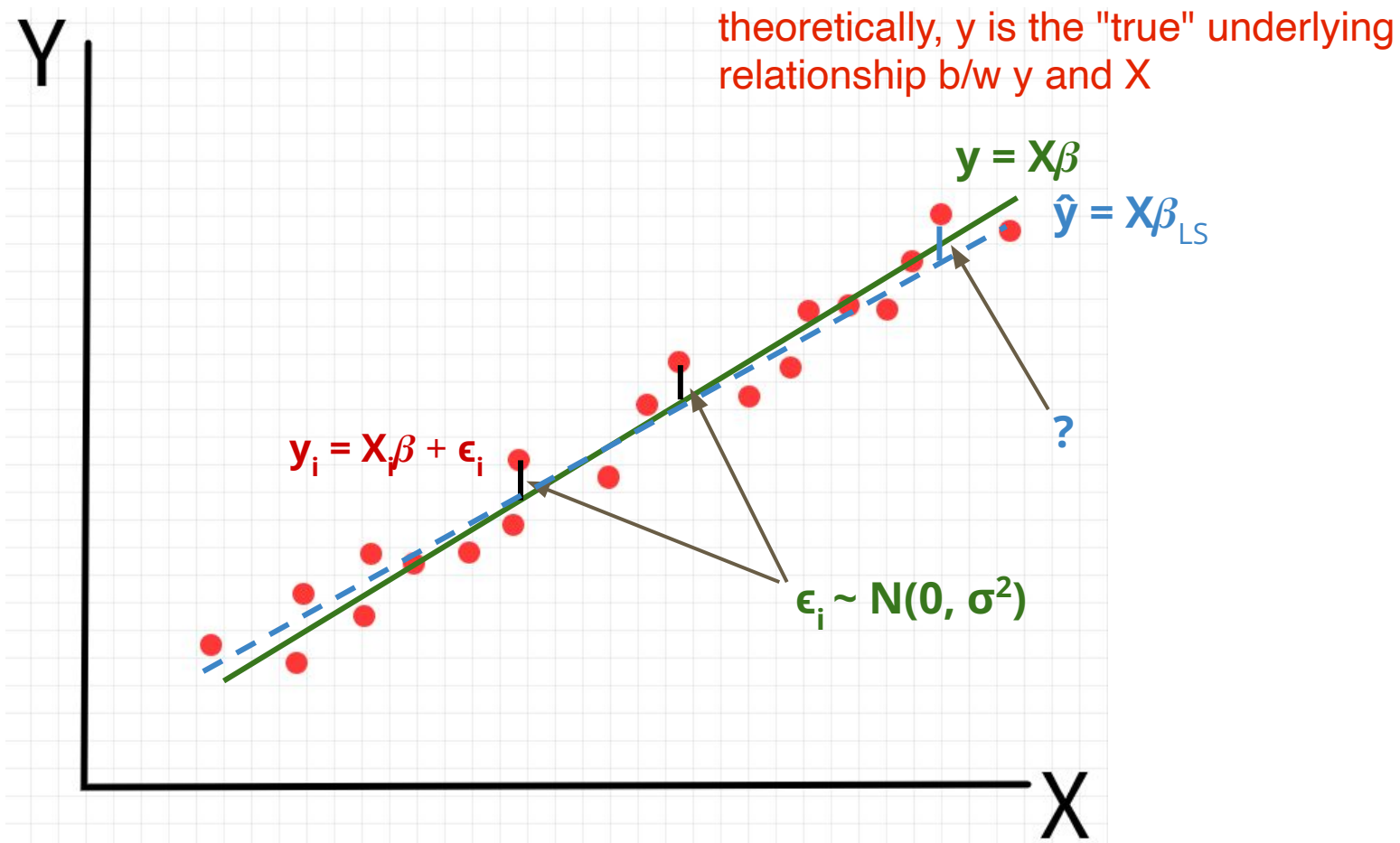


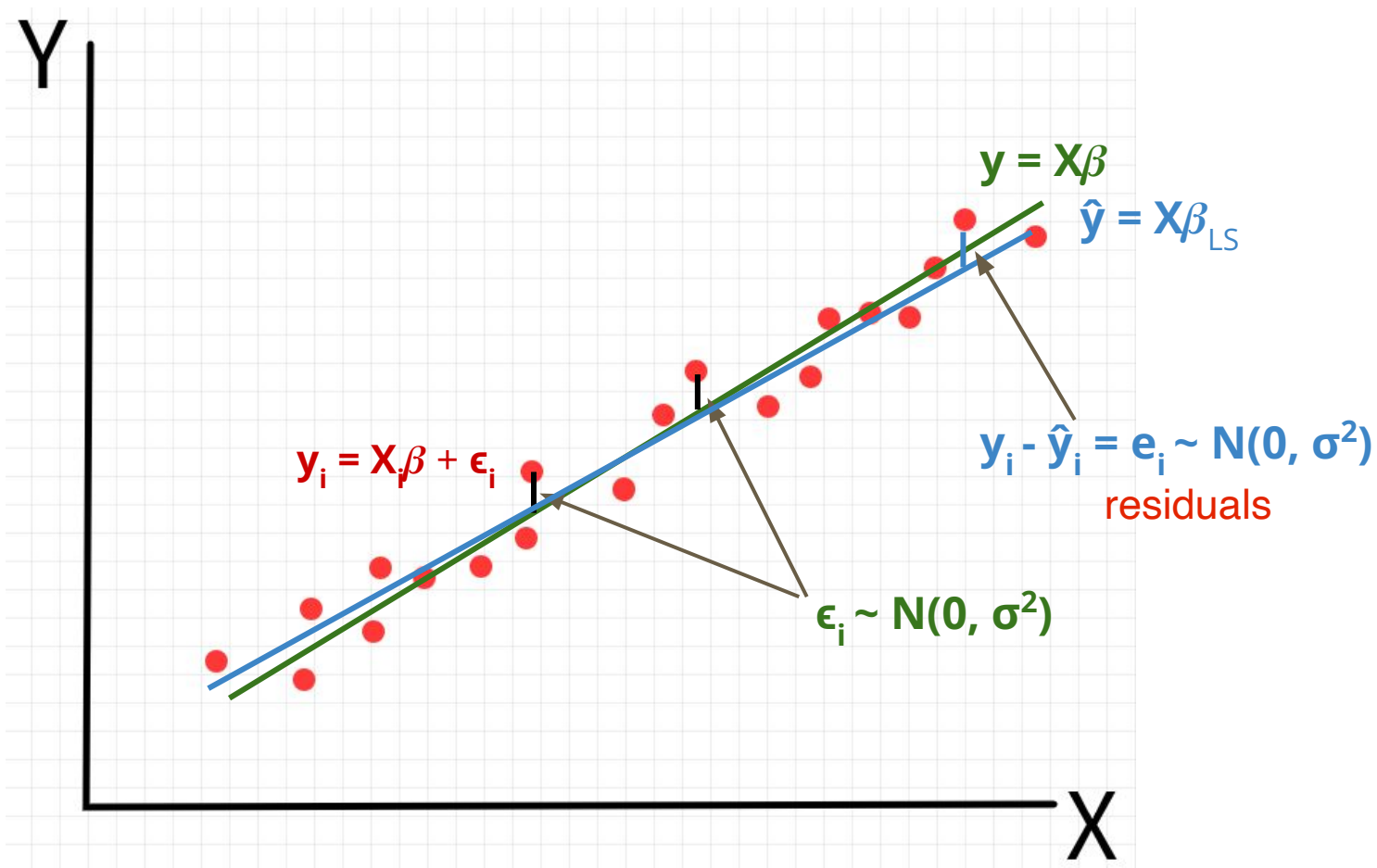


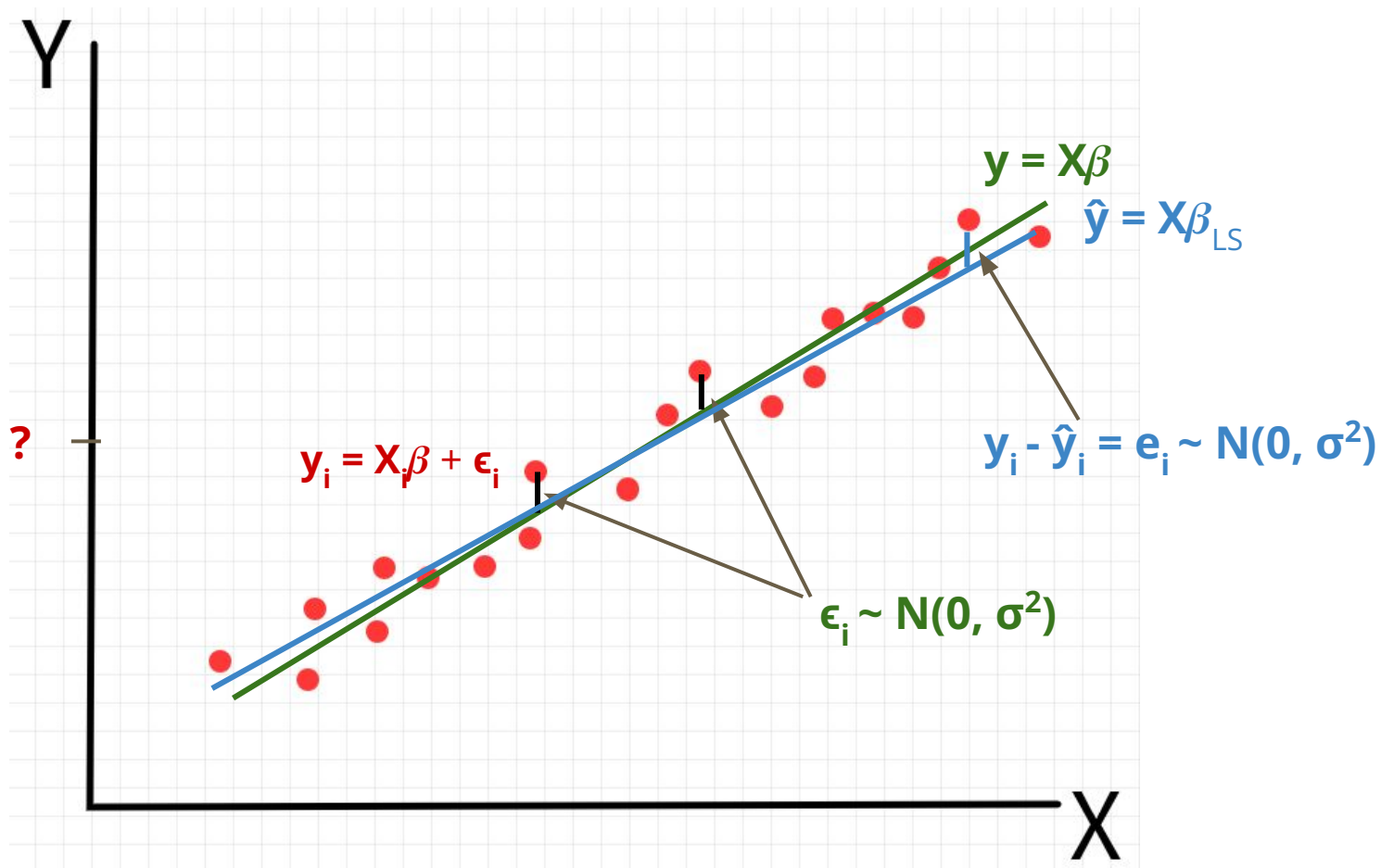


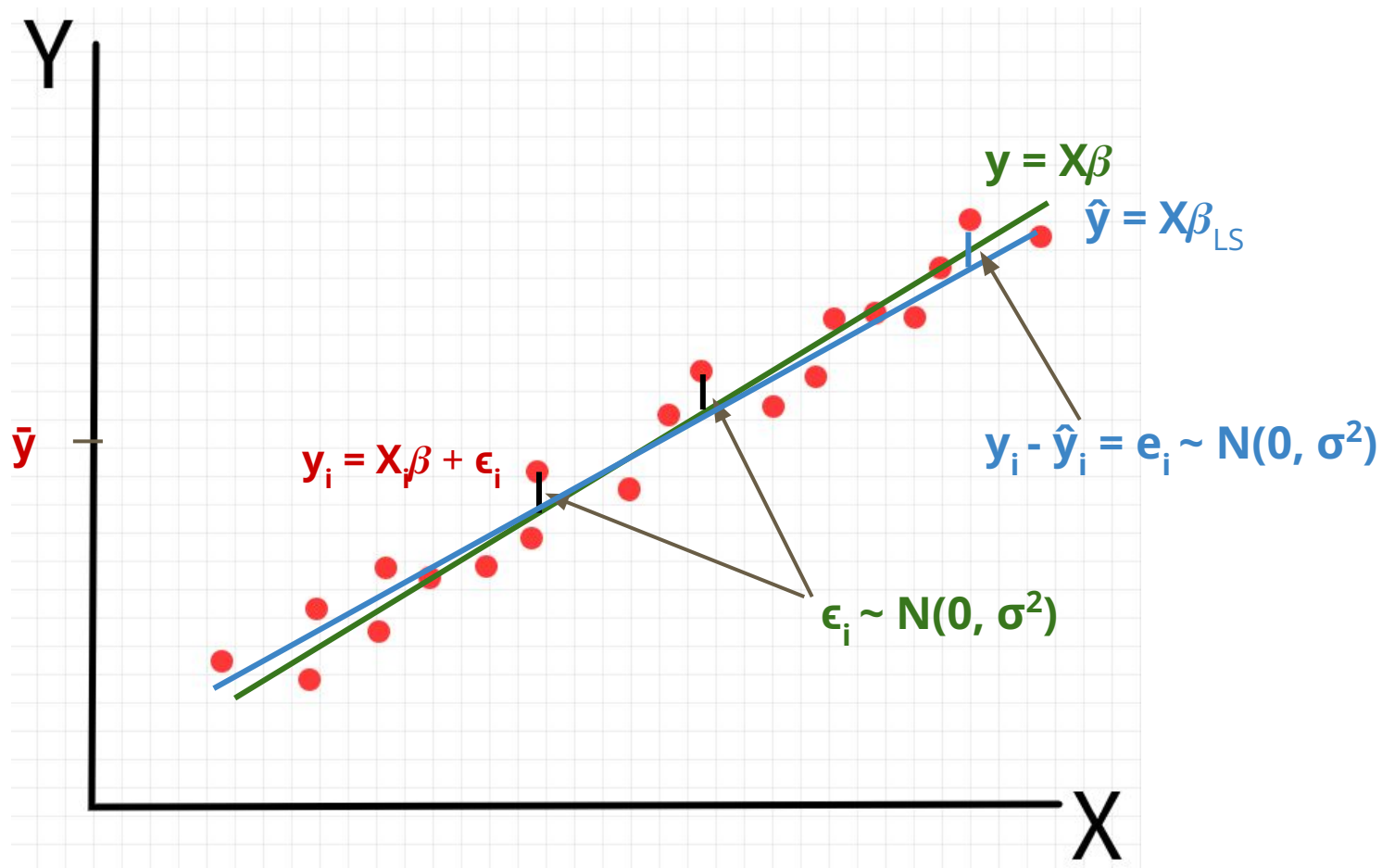












How do we know a linear model is applicable?

Is it true that $\mathbf{y} = \mathbf{x}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathbf{N}(\mathbf{0}, \sigma^2)$?

How do we know a linear model is applicable?

Is it true that $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathbf{N}(\mathbf{0}, \sigma^2)$?

If this is true then: $\mathbf{y}_i - \hat{\mathbf{y}}_i = \mathbf{e}_i \sim \mathbf{N}(0, \sigma^2)$

Can we check this?

- it's easier to check if it's not than if it is

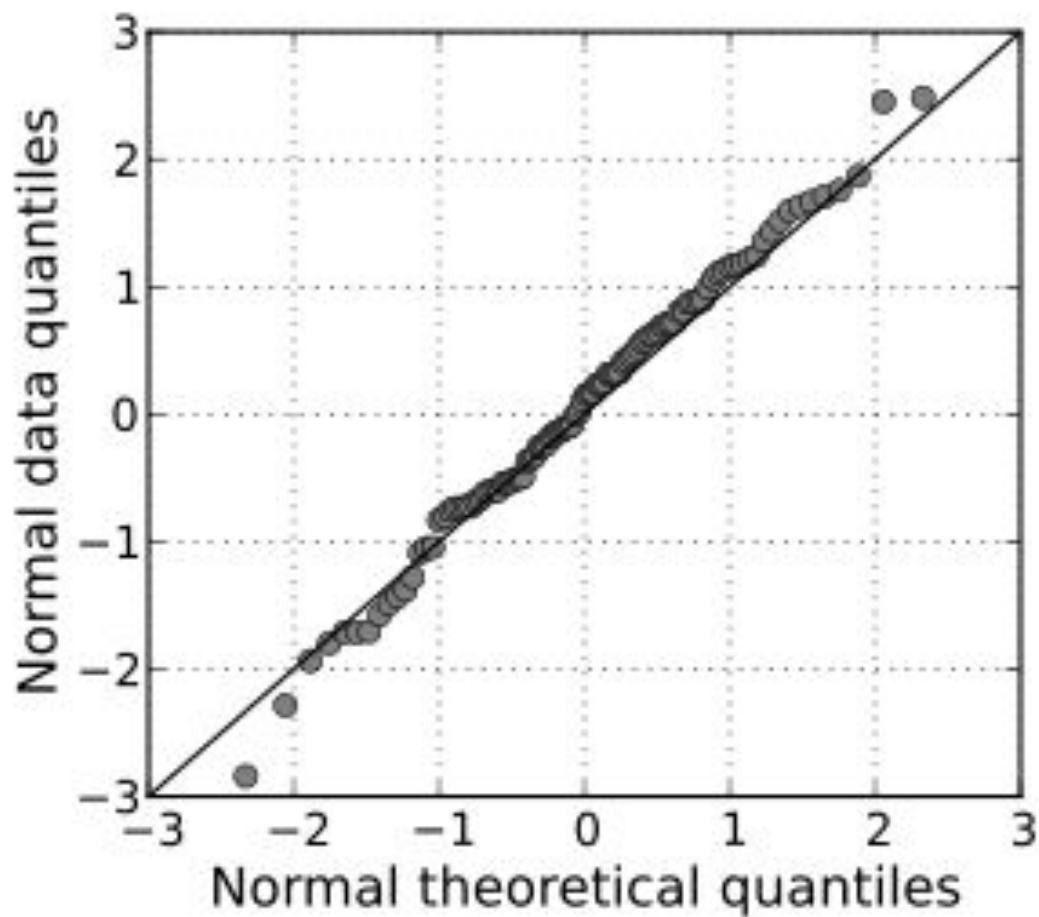
QQ plot (Checking the Normal assumption)

Quantiles are the values for which a particular % of values are contained below it.

For example the 50% quantile of a $N(0,1)$ distribution is 0 since 50% of samples would be contained below 0 were you to sample a large number of times.

For all quantiles q , if $\text{sample}.q == \text{known_distribution}.q$ then they have the same distribution.

QQ plot



demo

Evaluating Our Regression Model

Some Notation:

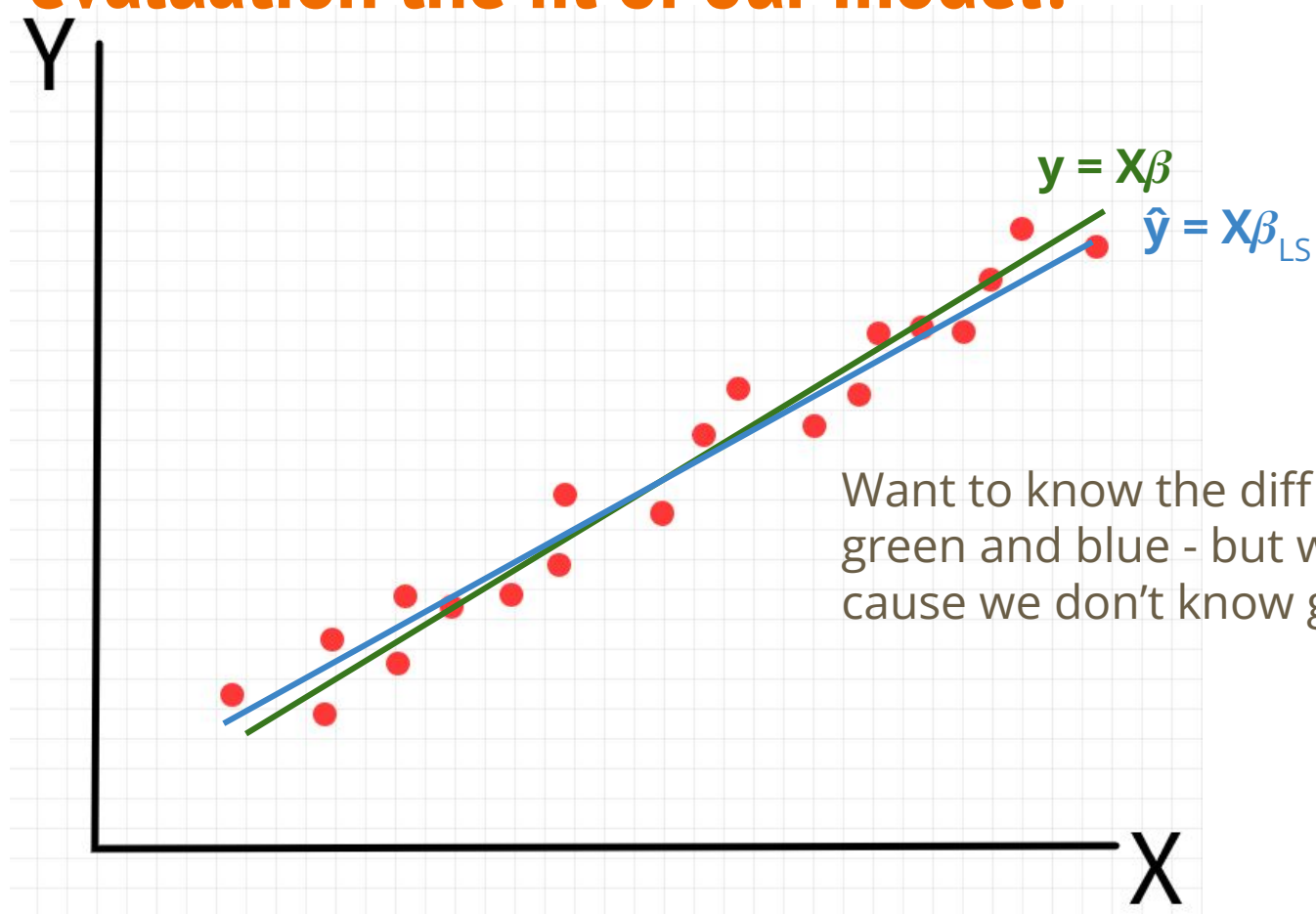
$\mathbf{y}_i$ is the “true” value from our data set (i.e. $\mathbf{x}_i\boldsymbol{\beta} + \boldsymbol{\epsilon}_i$)

$\hat{\mathbf{y}}_i$ is the estimate of y_i from our model (i.e. $\mathbf{x}_i\boldsymbol{\beta}_{LS}$)

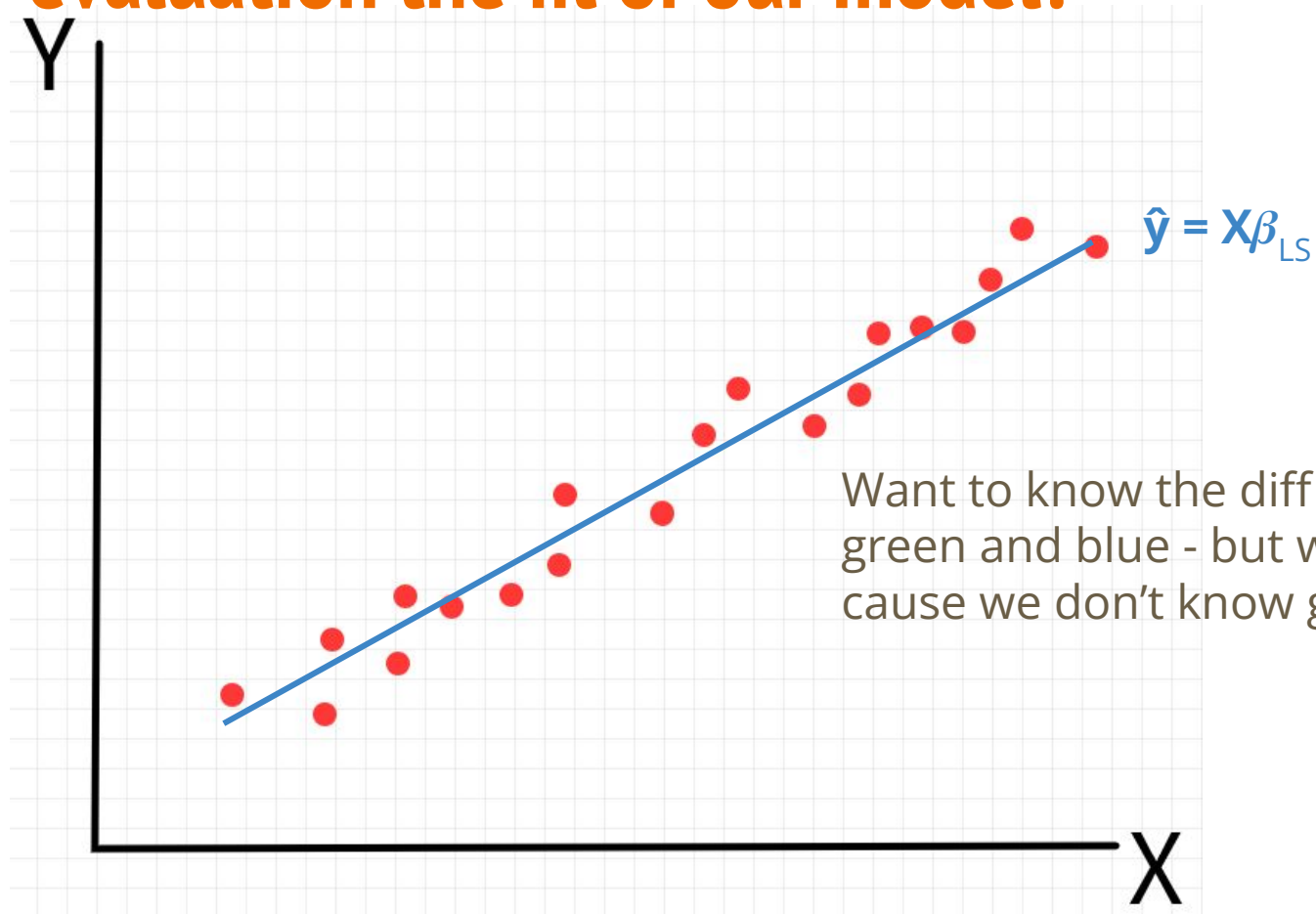
$\bar{\mathbf{y}}$ is the sample mean all $\mathbf{y}_i$

$\mathbf{y}_i - \hat{\mathbf{y}}_i$ are the estimates of $\boldsymbol{\epsilon}_i$ and are referred to as residuals

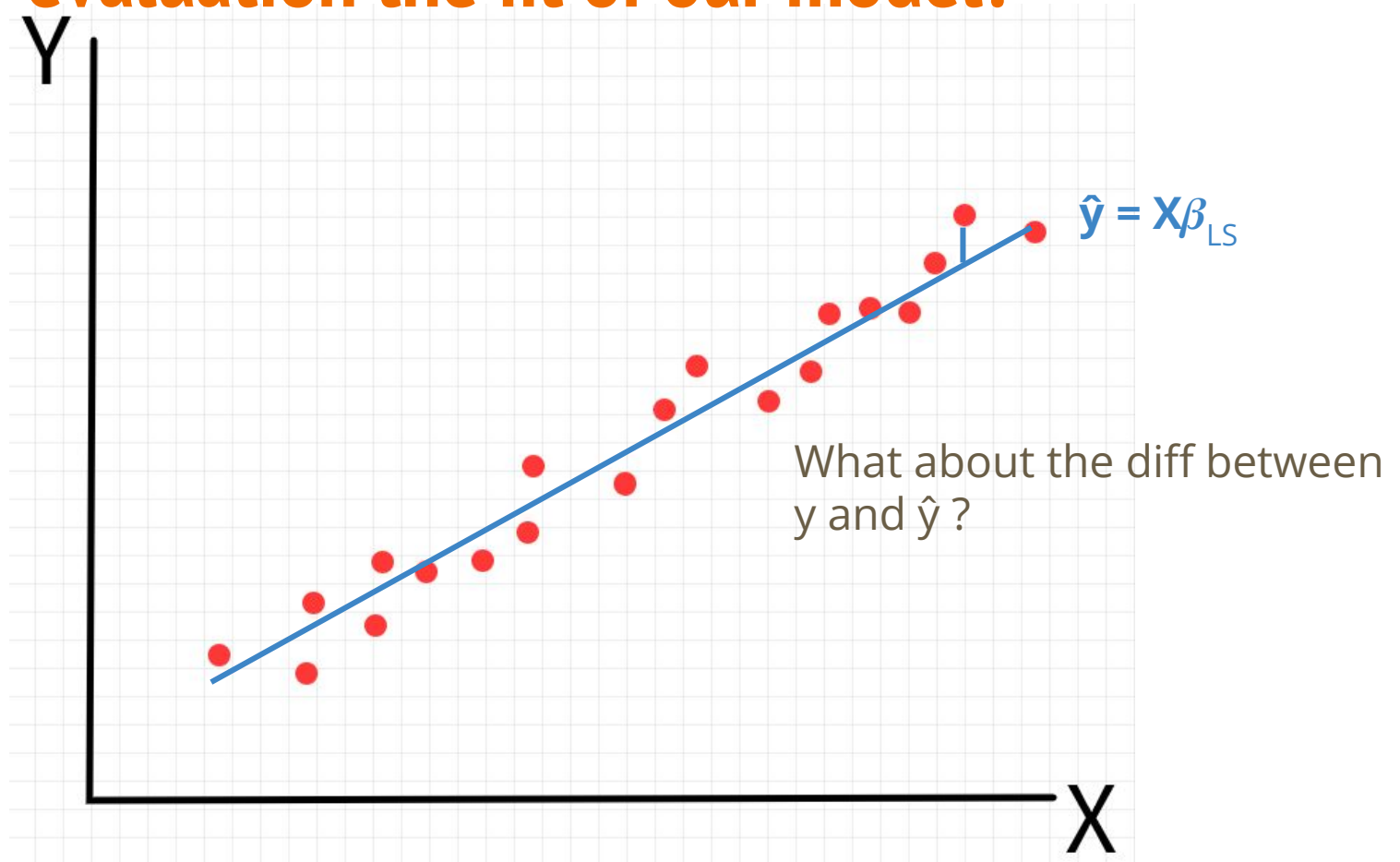
Metric for evaluation the fit of our model?



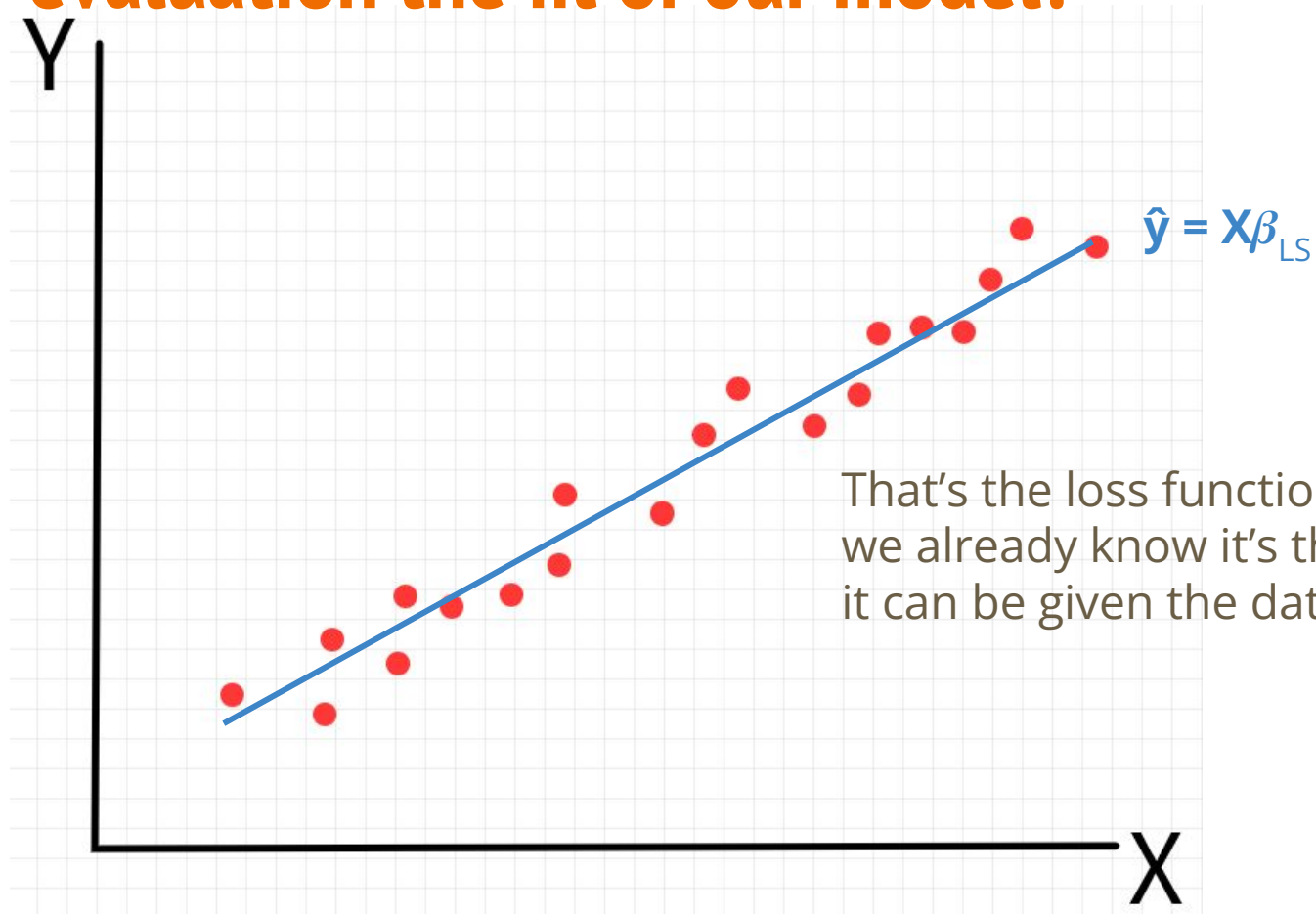
Metric for evaluation the fit of our model?



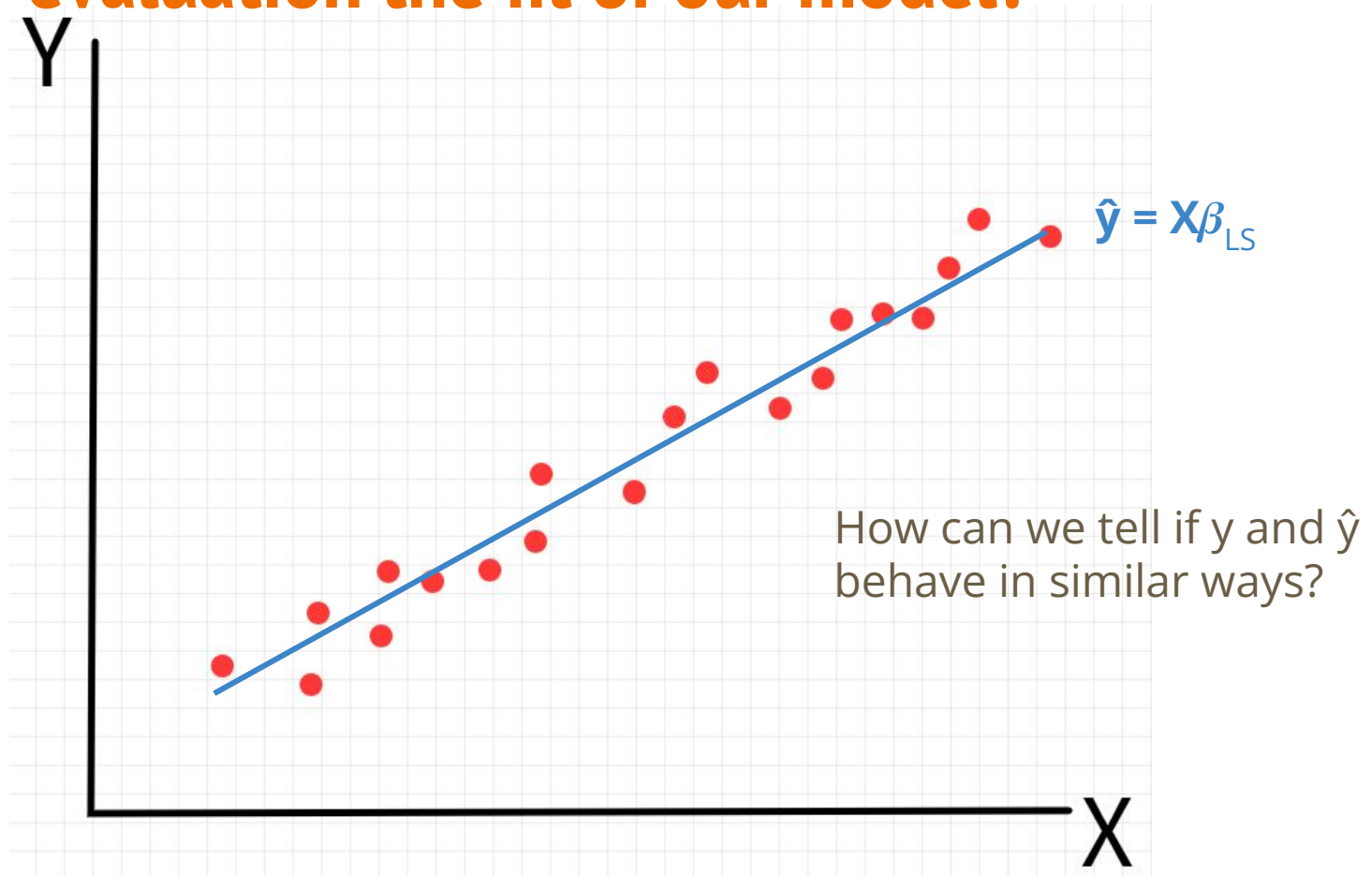
Metric for evaluation the fit of our model?



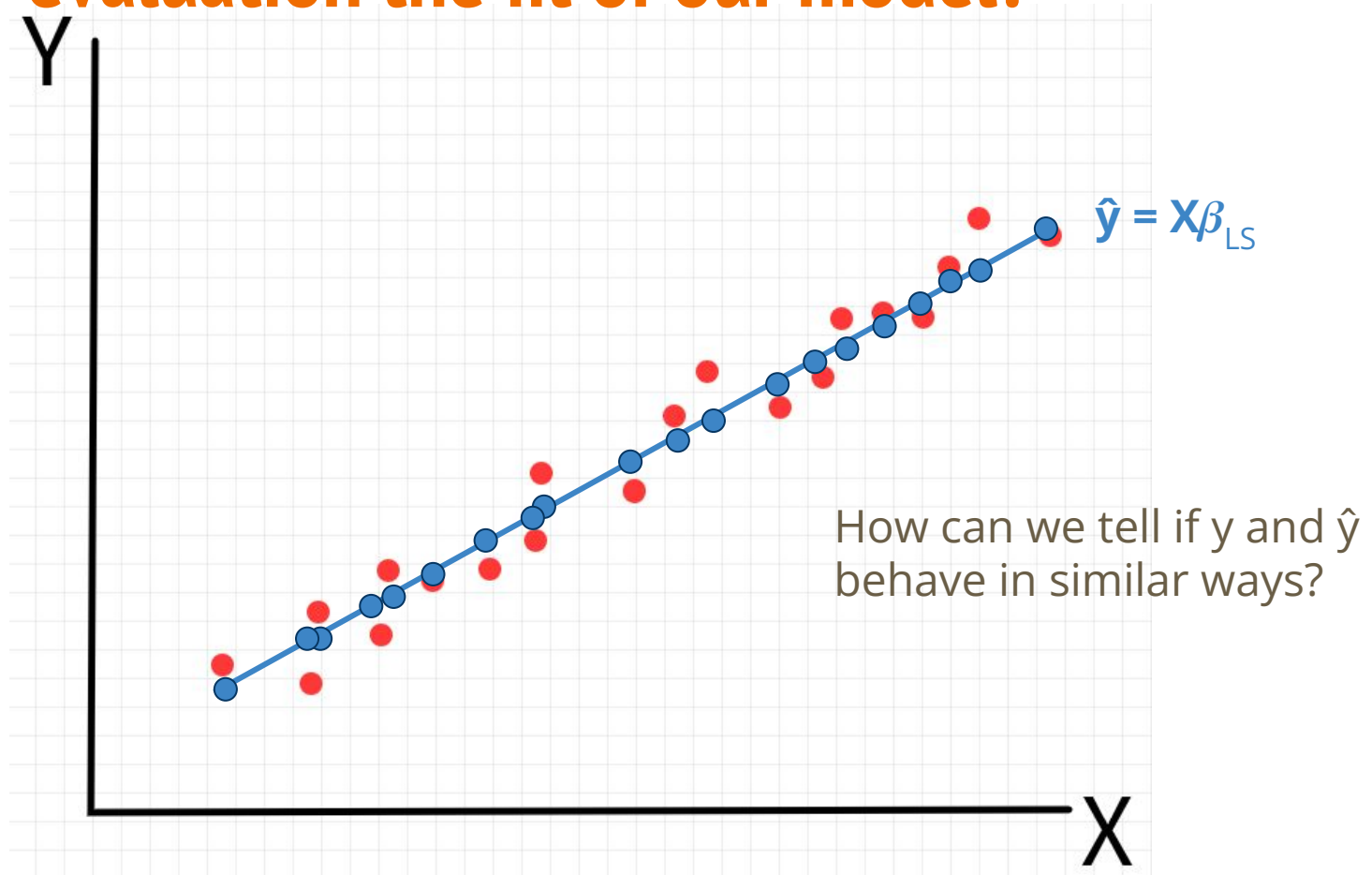
Metric for evaluation the fit of our model?



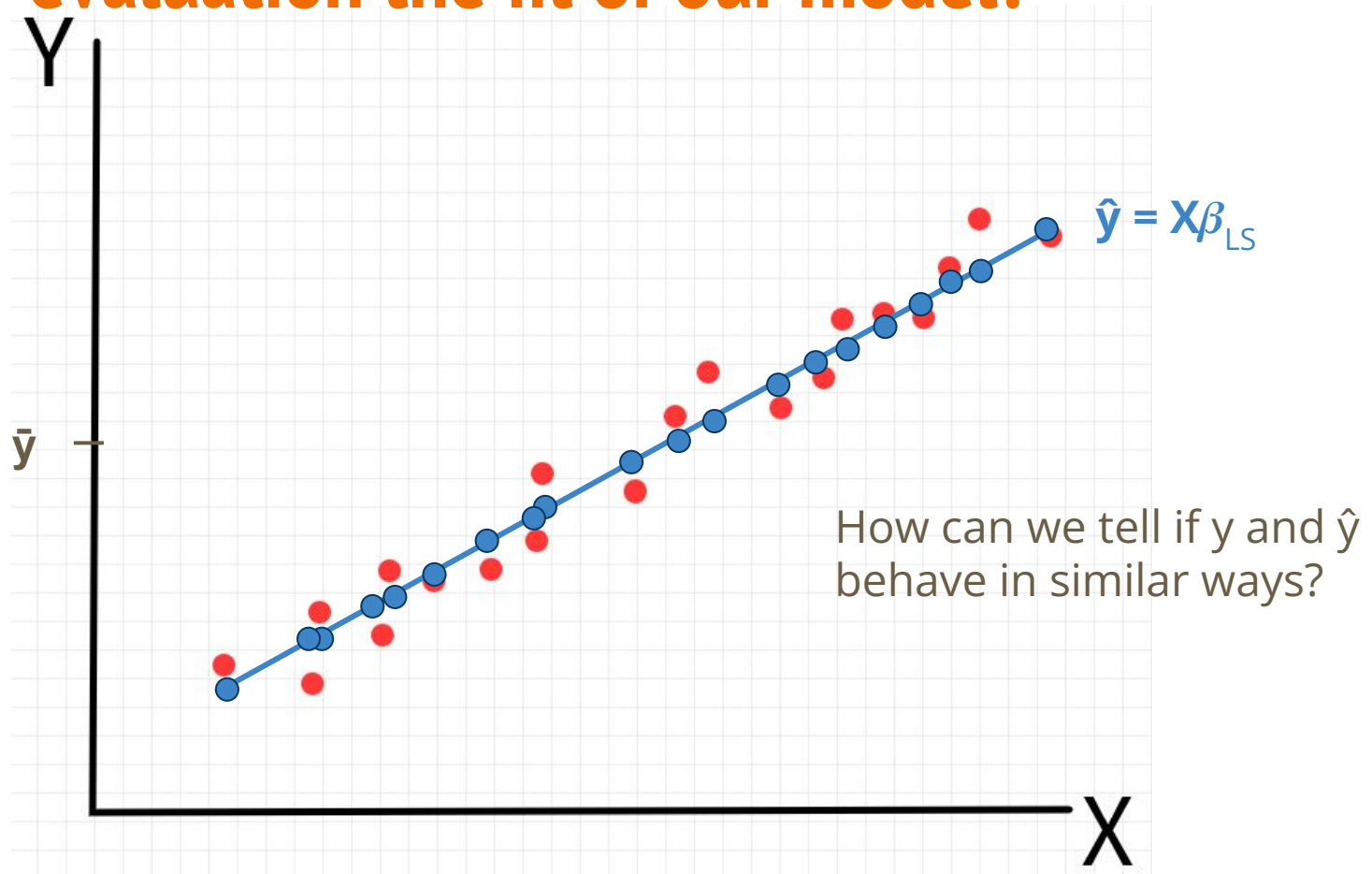
Metric for evaluation the fit of our model?



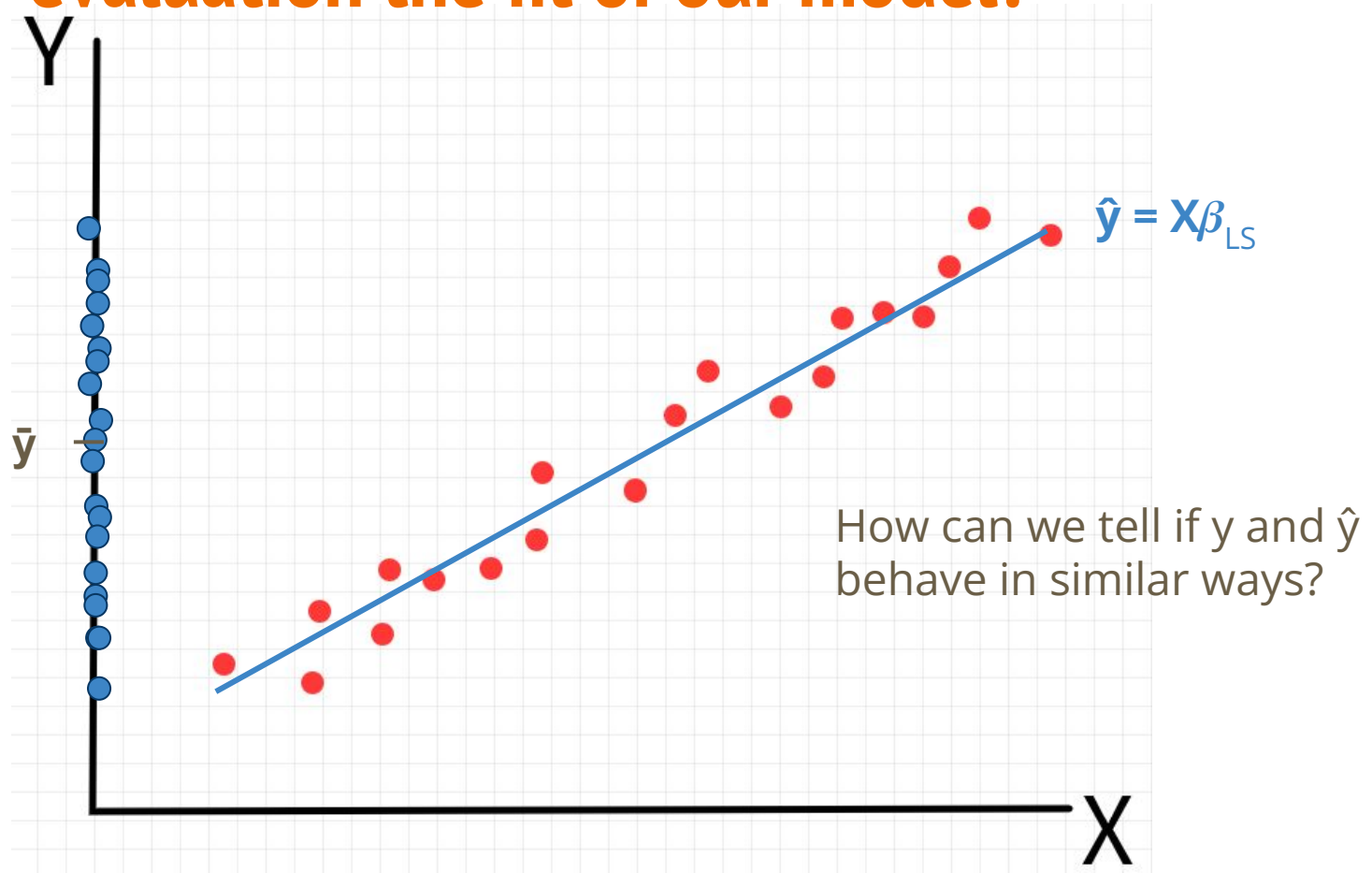
Metric for evaluation the fit of our model?



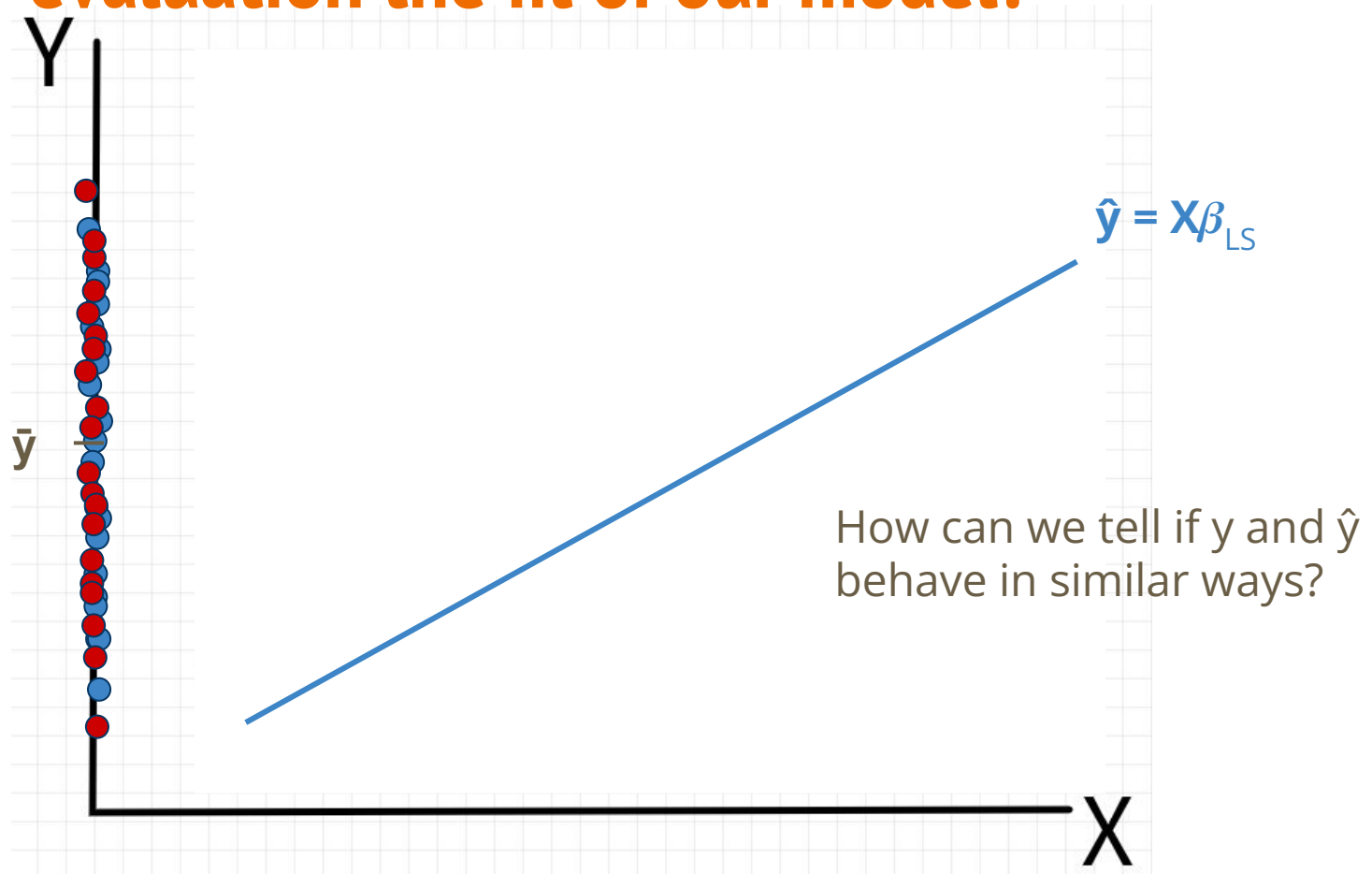
Metric for evaluation the fit of our model?



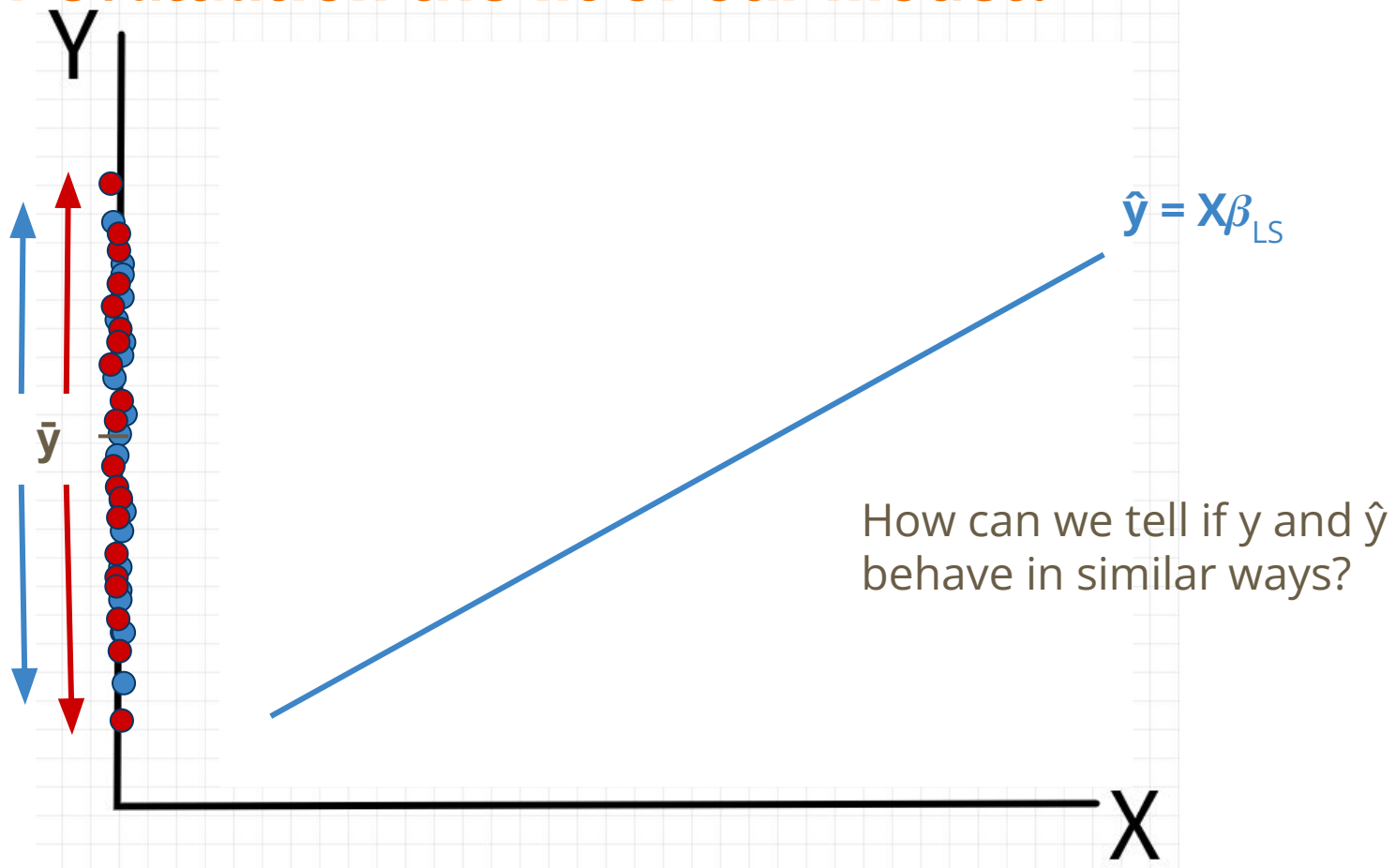
Metric for evaluation the fit of our model?



Metric for evaluation the fit of our model?



Metric for evaluation the fit of our model?



Metric for evaluation the fit of our model?

Is the value of the loss function sufficient? i.e.

$$\|y - X\beta\|_2^2 = \sum_i^n (y_i - \hat{y}_i)^2$$

Evaluating Our Regression Model

$$TSS = \sum_i^n (y_i - \bar{y})^2$$

← This is a measure of the spread of y_i around the mean of y

Evaluating Our Regression Model

$$TSS = \sum_i^n (y_i - \bar{y})^2$$

← This is a measure of the spread of y_i around the mean of y

$$ESS = \sum_i^n (\hat{y}_i - \bar{y})^2$$

← This is a measure of the spread of our model's estimates of y_i around the mean of y

Evaluating Our Regression Model

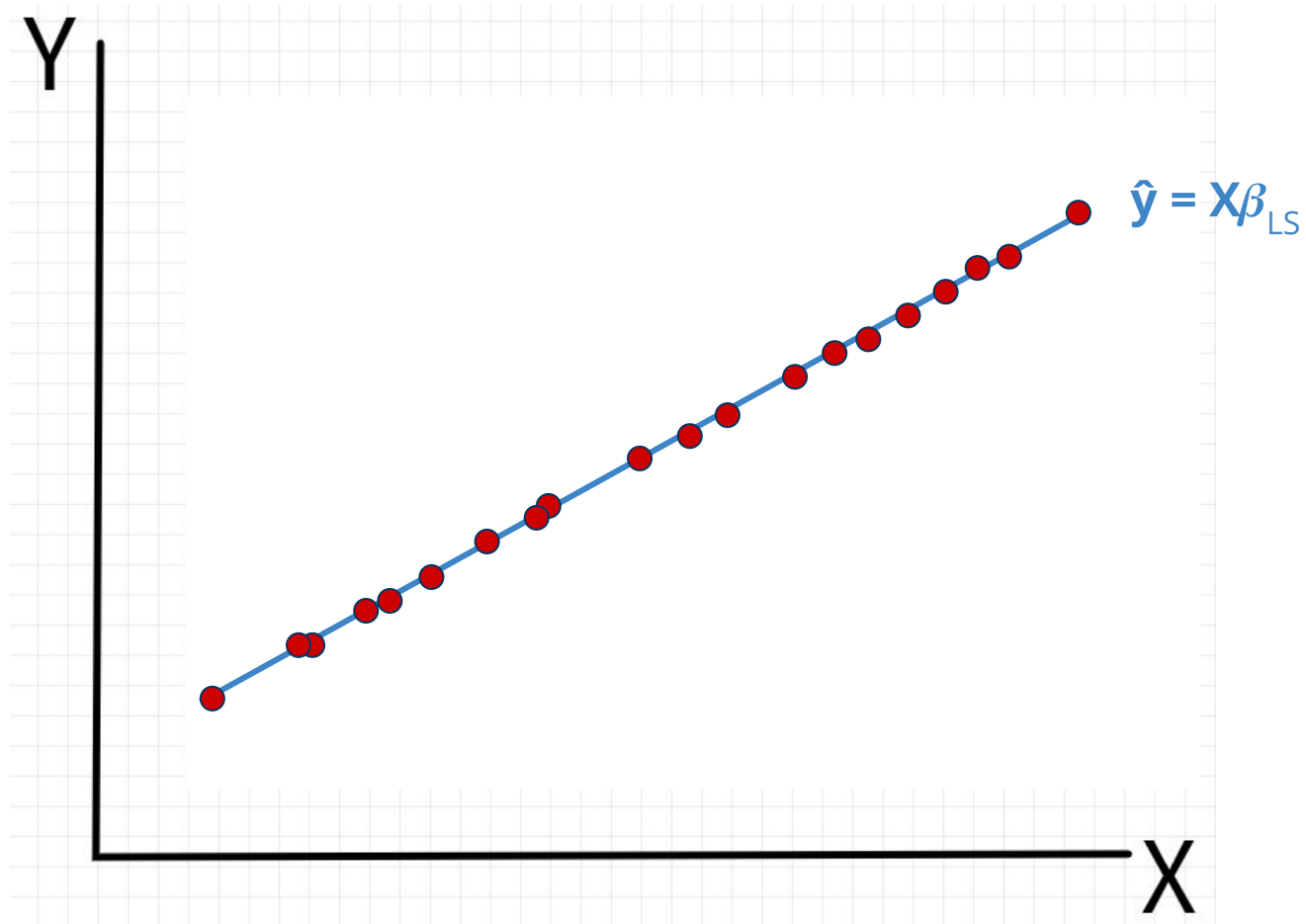
$$TSS = \sum_i^n (y_i - \bar{y})^2$$

$$R^2 = \frac{ESS}{TSS}$$

$$ESS = \sum_i^n (\hat{y}_i - \bar{y})^2$$

R^2 measures the fraction of variance that is explained by $\hat{y}$ (our model)

$$R^2 = 1$$



$$R^2 = 0$$



$$R^2 = 0$$



Evaluating our Regression Model

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.840			
Model:	OLS	Adj. R-squared:	0.836			
Method:	Least Squares	F-statistic:	254.1			
Date:	Sun, 20 Mar 2022	Prob (F-statistic):	2.72e-39			
Time:	11:36:16	Log-Likelihood:	-482.37			
No. Observations:	100	AIC:	970.7			
Df Residuals:	97	BIC:	978.5			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	2.1912	3.162	0.693	0.490	-4.085	8.467
x1	29.3912	3.274	8.977	0.000	22.893	35.889
x2	78.1391	3.594	21.741	0.000	71.006	85.272
=====						
Omnibus:	1.279	Durbin-Watson:	1.824			
Prob(Omnibus):	0.527	Jarque-Bera (JB):	1.065			
Skew:	0.253	Prob(JB):	0.587			
Kurtosis:	2.999	Cond. No.	1.38			
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Evaluating our Regression Model

```

=====
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```

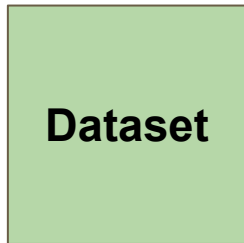
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```

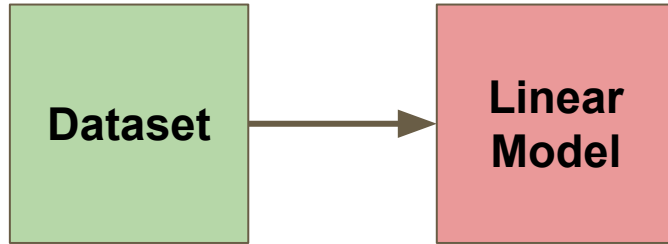
=====
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Prob(Omnibus):        0.527    Jarque-Bera (JB):        1.065
Skew:                 0.253    Prob(JB):                0.587
Kurtosis:             2.999    Cond. No.:               1.38
=====

```

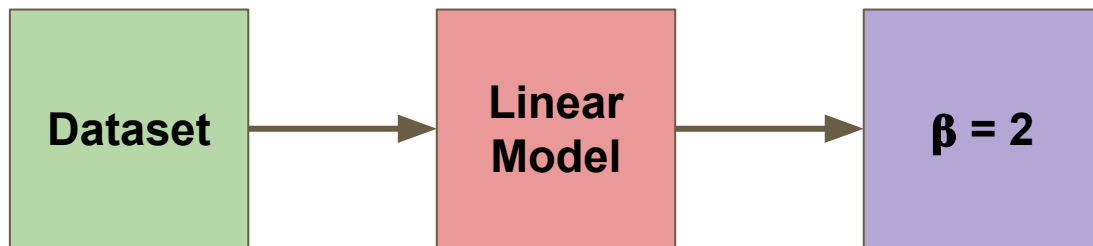
Hypothesis Testing



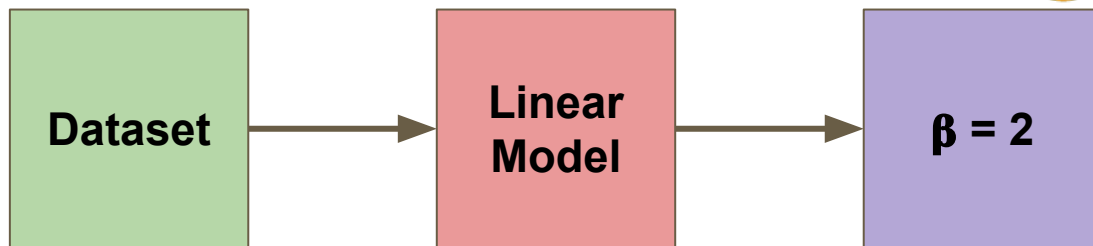
Hypothesis Testing



Hypothesis Testing



Hypothesis Testing



Could the
real beta
be 5?

HHHHHHHH



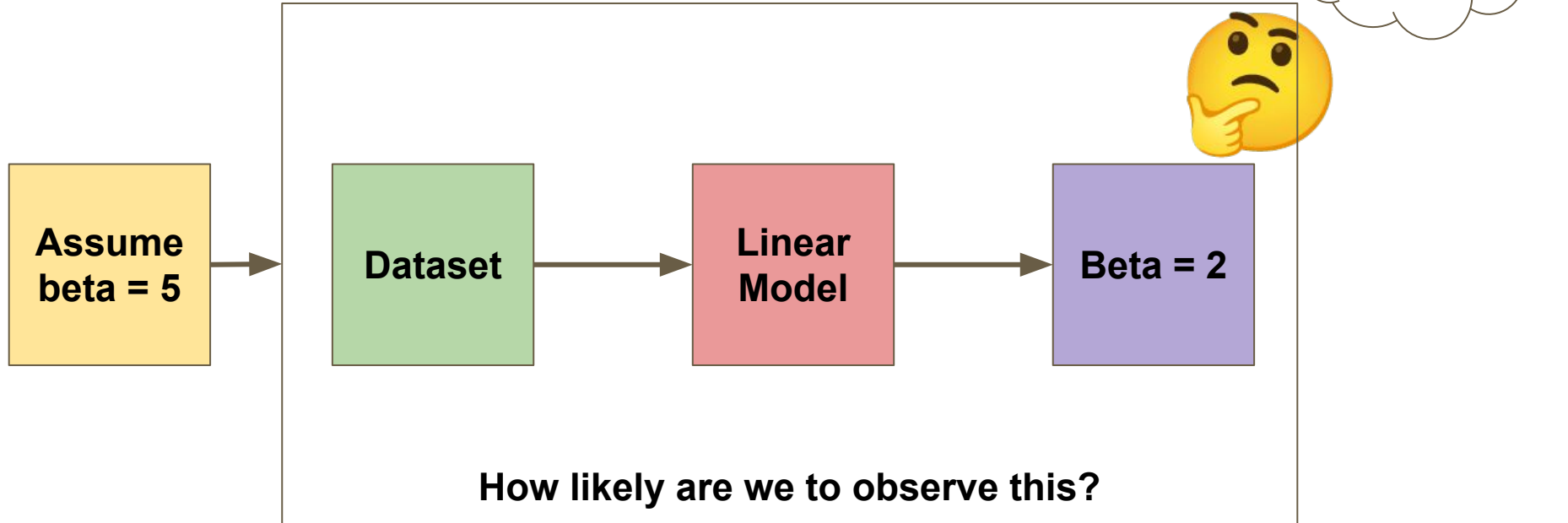
The coin is
probably
not fair

HTHTHTTT

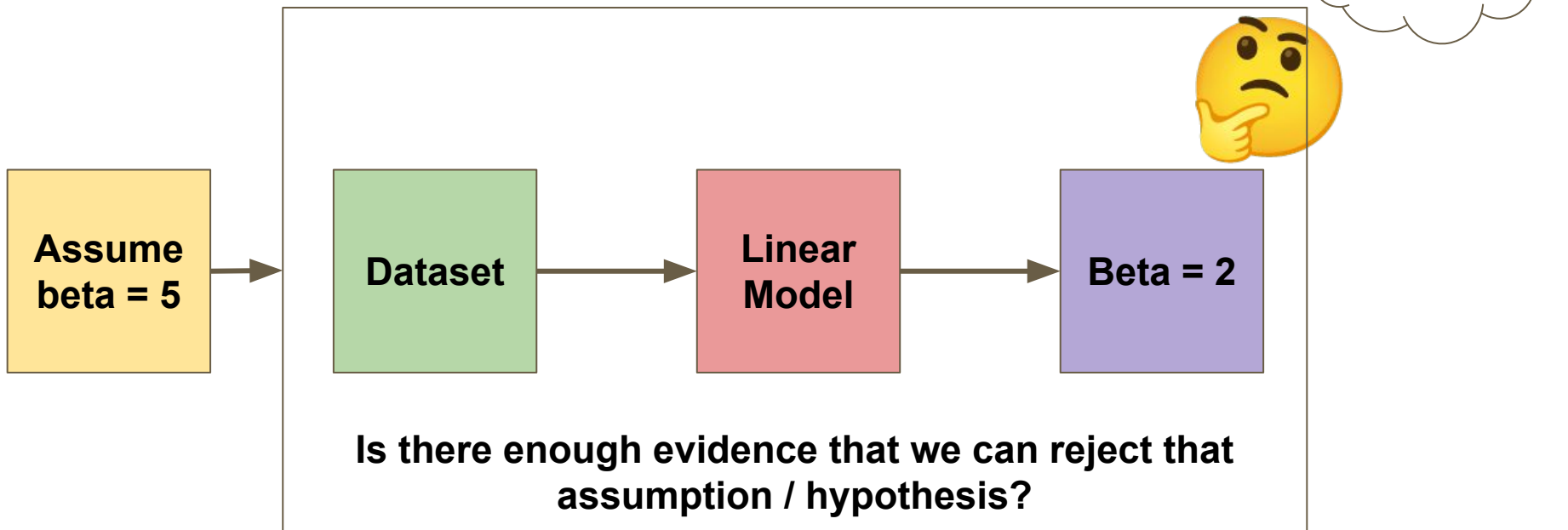


The coin
could be
fair

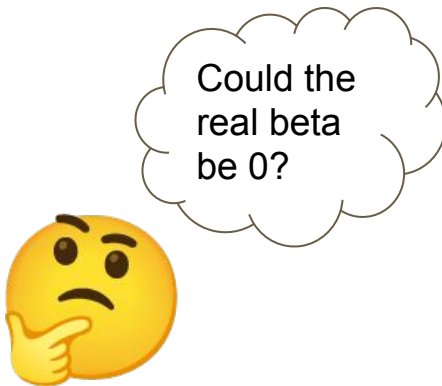
Hypothesis Testing



Hypothesis Testing



worksheet



Hypothesis Testing

Each parameter of an independent variable $\mathbf{x}$ has an associated confidence interval and t-value + p-value.

If the parameter / coefficient is not significantly distinguishable from 0 then we cannot assume that there is a significant linear relationship between that independent variable and the observations $\mathbf{y}$ (i.e. if the interval includes 0 or if the p-value is too large)

Hypothesis Test

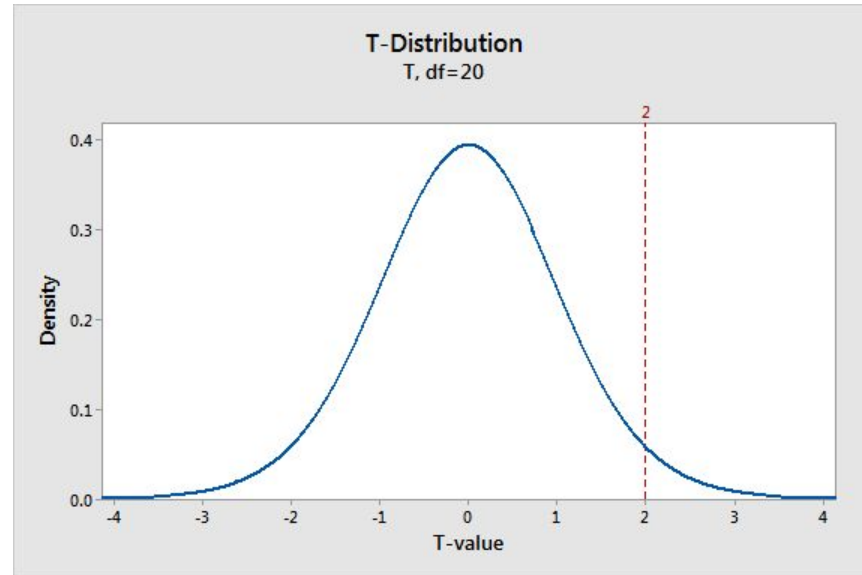
We want to know if there is evidence to reject the hypothesis $H_0 : \beta = 0$ (i.e. that there is no linear relation between X and Y) using the information from $\hat{\beta}$.

We want to know the largest probability of obtaining the data observed, under the assumption that the null hypothesis is correct.

How do we obtain that probability?

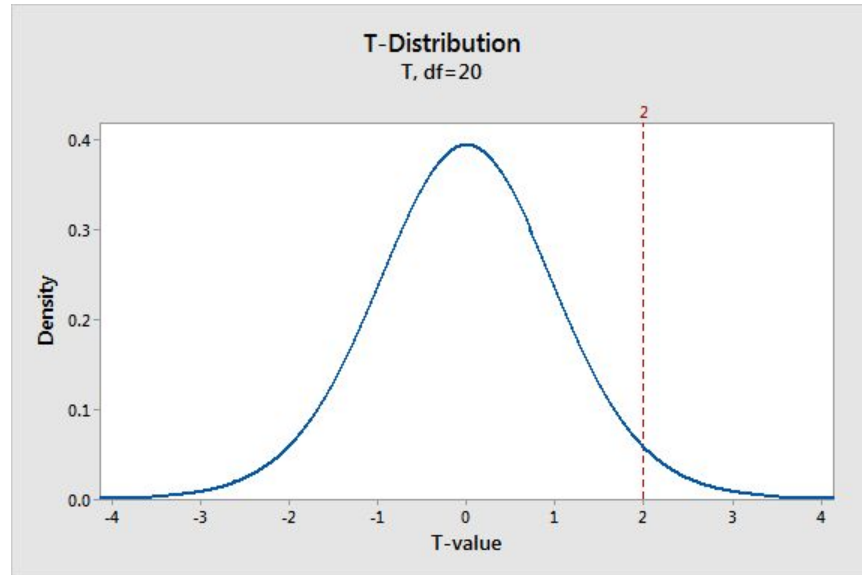
Hypothesis Test

Under the null hypothesis what should be the distribution of the normalized estimates? T-distribution (parametrized by the sample size)



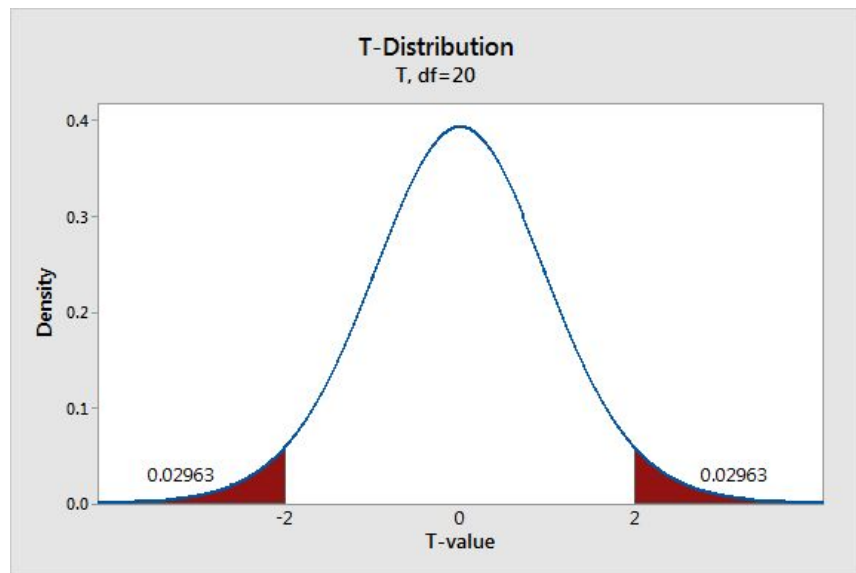
Hypothesis Test

We can then compute the t-value that corresponds to the sample we observed.



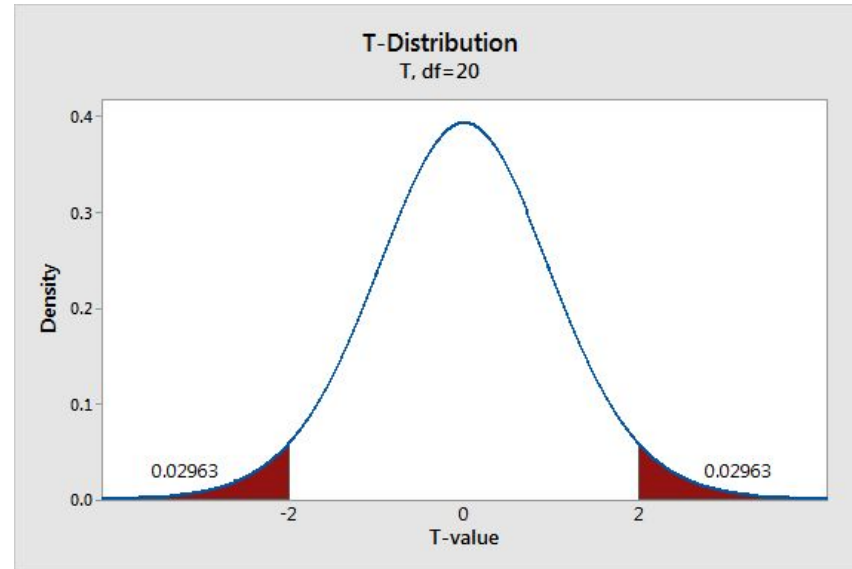
Hypothesis Test

And then compute the probability of observing estimates of β at least as extreme as the one observed. (i.e. trying to find evidence against H_0)



Hypothesis Test

This probability is called a p-value.



Hypothesis Test

A p-value **smaller than a given threshold** would mean the data was unlikely to be observed under H_0 so we can reject the hypothesis H_0 . If not, then we lack the evidence to reject H_0 .

	coef	std err	t	P> t	[0.025	0.975]
const	2.1912	3.162	0.693	0.490	-4.085	8.467
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x2	78.1391	3.594	21.741	0.000	71.006	85.272

Hypothesis Test

Which parameters should we not include in our linear model?

	coef	std err	t	P> t	[0.025	0.975]
const	2.1912	3.162	0.693	0.490	-4.085	8.467
x1	29.3912	3.274	8.977	0.000	22.893	35.889
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Skew:	0.253	Prob(JB):	0.587			
Kurtosis:	2.999	Cond. No.	1.38			

Confidence Intervals

Goal: for a given confidence level (let's say 90%), construct an interval around an estimate such that, if the estimation process were repeated indefinitely, the interval would contain the true value (that the estimate is estimating) 90% of the time.

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x2	78.1391	3.594	21.741	0.000	71.006	85.272

Z-values

These are the number of standard deviations from the mean of a $N(0,1)$ distribution required in order to contain a specific % of values were you to sample a large number of times.

To find the .95 z-value (the value z such that 95% of the observations lie within z standard deviations of the mean ($\mu \pm z * \sigma$)) you need to solve:

$$\int_{-z}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = .95$$

Z-values

The .95 z-value is 1.96.

This means 95% of observations from a $N(\mu, \sigma)$ lie within 1.96 standard deviations of the mean ($\mu \pm 1.960 * \sigma$)

If we get a sample from a $N(\mu, \sigma)$ of size n , how would we create a confidence interval around the estimated mean?

Confidence Intervals

How do we build a confidence interval?

Assume $\mathbf{Y}_i \sim \mathbf{N}(5, 25)$, for $1 \leq i \leq 100$ and $\mathbf{y}_i = \mu + \epsilon$ where $\epsilon \sim \mathbf{N}(0, 25)$. Then the Least Squares estimator of μ (μ_{LS}) is

the sample mean $\bar{y}$

What is the 95% confidence interval for μ_{LS} ?

$$\begin{aligned} CI_{.95} &= [\bar{y} - 1.96 \times SE(\mu_{LS}), \bar{y} + 1.96 \times SE(\mu_{LS})] \\ &= [\bar{y} - 1.96 \times .5, \bar{y} + 1.96 \times .5] \end{aligned}$$

$$\begin{aligned} SE(\mu_{LS}) &= \sigma_{\epsilon} / \sqrt{n} \\ &= 5 / \sqrt{100} \\ &= .5 \end{aligned}$$

Z-value for 95% Confidence Interval

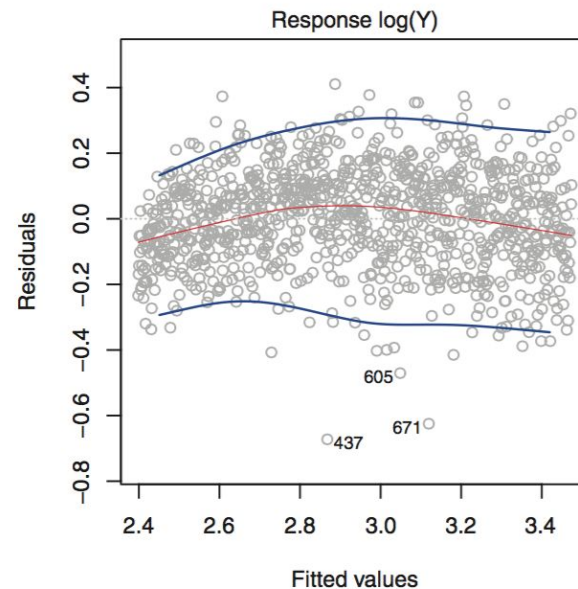
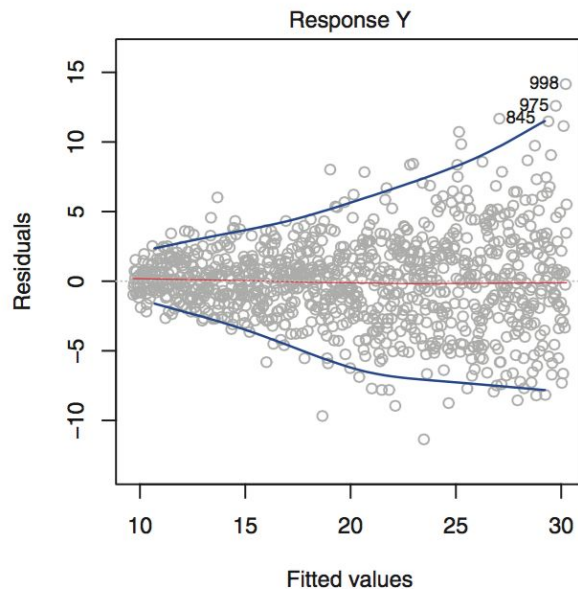
Checking our Assumptions

1. Normal Distribution?
2. Constant Variance?

Constant Variance

One of our assumptions was that our noise had constant variance. How can we verify this?

We can plot residuals (noise estimates) for each fitted value $\hat{y}_i$



Extending our Linear Model

Changing the assumptions we made can drastically change the problem we are solving. A few ways to extend the linear model:

1. Non-constant variance - used in WLS (weighted least squares)
2. Distribution of error is not Normal - used in GLM (generalized linear models)